



**SANITARY LANDFILL ENVIRONMENTAL MONITORING PLAN
(SLEMP)**

**HAPPY JACK LANDFILL
CHEYENNE, WYOMING**

April 12, 2017

Revised August 23, 2017

Project #: 47A-001-001

PREPARED BY: Trihydro Corporation

1252 Commerce Drive, Laramie, WY 82070



ENGINEERING SOLUTIONS. ADVANCING BUSINESS.

Home Office | 1252 Commerce Drive | Laramie, WY 82070 | phone 307/745.7474 | fax 307/745.7729 | www.trihydro.com

Table of Contents

1.0	INTRODUCTION	1-1
2.0	GROUNDWATER AND LEACHATE MONITORING PLAN	2-1
2.1	Phases of Groundwater Monitoring	2-1
2.1.1	Detection Monitoring	2-1
2.1.2	Assessment Monitoring	2-2
2.2	Groundwater Bearing Zones, Aquifers and Leachate Monitoring Points	2-2
2.2.1	Perched 1 Water Bearing Zone	2-3
2.2.2	Perched 2 Water Bearing Zone	2-4
2.2.3	Principal Unconfined Aquifer	2-4
2.2.4	Semi-Confined Aquifer.....	2-5
2.2.5	Leachate Wells	2-6
2.2.6	Leachate Sumps.....	2-6
2.3	Groundwater Monitoring Procedures	2-7
2.3.1	Fluid Level Monitoring.....	2-7
2.3.2	Well Evacuation Procedures	2-8
2.3.2.1	Well Evacuation Using a Dedicated Pump	2-8
2.3.2.2	Groundwater Evacuation Using Bailers	2-8
2.3.2.3	Alternative Sampling Techniques	2-9
2.3.3	Field Analyses.....	2-9
2.3.4	Sample Withdrawal.....	2-9
2.3.5	Field Measurements, Calibration, and Decontamination	2-10
2.3.6	Documentation Requirements	2-11
2.3.7	Chain-of-Custody	2-11
2.3.8	Custody for Shipment	2-12
2.3.9	Sample Shipment Procedures.....	2-12
2.3.10	System Audits	2-13
2.3.11	Corrective Measures.....	2-13
2.4	Future Expansion Area Wells.....	2-13
2.5	Off-Site Water Well Monitoring Program.....	2-14
2.5.1	Baseline Monitoring Program	2-14
2.5.2	Expanded Monitoring Program	2-15

Table of Contents (cont.)

3.0	GROUNDWATER DATA QUALITY ASSURANCE/QUALITY-CONTROL	3-1
3.1	Quality Control Parameters	3-1
3.2	Quality Assurance and Quality Control Measures	3-2
3.2.1	Precision	3-3
3.2.2	Accuracy	3-3
3.2.3	Method Compliance	3-4
3.2.4	Completeness	3-4
4.0	GROUNDWATER DATA STATISTICAL ANALYSES	4-1
4.1	General Statistics	4-1
4.1.1	Non-detect Data	4-2
4.2	Interwell Statistics	4-2
4.3	Intrawell Statistics	4-2
4.4	Comparisons to Groundwater Protection Standards	4-3
4.4.1	Tolerance Limit Analysis	4-3
5.0	LANDFILL METHANE MONITORING PLAN	5-1
5.1	Methane Monitoring at the Landfill Boundary	5-1
5.2	Methane Monitoring of Equipment Buildings	5-1
6.0	NOTIFICATION REQUIREMENTS	6-1
6.1	Methane Monitoring Notification Requirements	6-1
6.2	Detection Monitoring Notification Requirements	6-1
6.3	Assessment Monitoring Notification Requirements	6-2
7.0	REPORTING REQUIREMENTS	7-1
7.1	Event Report	7-1
7.2	Annual Report	7-1
8.0	REFERENCES	8-1

List of Tables

1. Monitoring Well History, Happy Jack Landfill, Cheyenne, Wyoming
2. Current Monitoring Well, Sump, and Piezometer Construction Summary, Happy Jack Landfill, Cheyenne, Wyoming
3. Sampling Schedule Summary, Happy Jack Landfill, Cheyenne, Wyoming
4. Fluid Monitoring Schedule Summary, Happy Jack Landfill, Cheyenne, Wyoming
5. WDEQ/SHWD Appendix A Constituent List, Happy Jack Landfill, Cheyenne, Wyoming
6. WDEQ/SHWD Appendix B Constituent List, Happy Jack Landfill, Cheyenne, Wyoming
7. WDEQ/SHWD Appendix C Constituent List, Happy Jack Landfill, Cheyenne, Wyoming
8. Groundwater Wells Permitted for Domestic or Stock Watering Purposes within ½ Mile of Expansion Area, Happy Jack Landfill, Cheyenne, Wyoming
9. Background and Compliance Monitoring Wells for Interwell Statistical Comparisons, Happy Jack Landfill, Cheyenne, Wyoming

List of Figures

1. Monitoring Well Location Map, City of Cheyenne Sanitary Landfill, Cheyenne, Wyoming

List of Attachments

- A. BORING LOGS AND WELL CONSTRUCTION DIAGRAMS
- B. EXAMPLES OF FIELD FORMS

1.0 INTRODUCTION

This Sanitary Landfill Environmental Monitoring Plan (SLEMP) has been prepared to address groundwater, leachate, and methane monitoring requirements at the City of Cheyenne Sanitary Landfill, also known as the Happy Jack Landfill (Landfill). The plan has been prepared as part of the facility's 2017 lifetime permit application, primarily to fulfill monitoring requirements contained within the Wyoming Department of Environmental Quality/Solid and Hazardous Waste Division (WDEQ/SHWD) municipal solid waste landfill regulations. Additional monitoring requirements have also been incorporated pursuant to the WDEQ November 25, 2015 Location Standards Variance Approval for the expansion area. This SLEMP has been prepared in accordance with revised Chapter 2 of the WDEQ/SHWD Rules and Regulations dated January 3, 2017.

The primary objective of the environmental monitoring program at the Landfill is early identification of impacts to the subsurface as a result of landfill activities. Additionally, the hydrogeologic and analytical data generated during routine monitoring may be used for assessment and/or remedial design purposes, if impacts are identified in subsurface soils or groundwater beneath the landfill.

The remainder of the SLEMP is organized into the following sections:

- Section 2.0 – Groundwater and Leachate Monitoring Plan: provides a description for each of the aquifers beneath the landfill, leachate monitoring points, different phases of monitoring, frequency of sampling, as well as fluid gauging and groundwater sampling procedures. This section also addresses the off-site water well monitoring program for wells within a half mile of the expansion area boundary that are permitted or certificated for domestic or stock use.
- Section 3.0 – Groundwater Data Quality Assurance/Quality Control: presents the quality assurance objectives for this project and the methodology for reviewing field and laboratory data.
- Section 4.0 – Groundwater Data Statistical Analyses: includes a description of the statistical methods used to evaluate the groundwater analytical results from each monitoring event.
- Section 5.0 – Landfill Methane Monitoring Plan: describes the monitoring system, frequency, and procedures for monitoring methane in landfill structures and in the subsurface.
- Section 6.0 – Notification Requirements: outlines the notification requirements and follow up actions to statistically significant increases in contaminant concentrations in groundwater, detections above groundwater protection standards, and methane concentration measuring more than 25% or 100% of the lower explosive limit (LEL).

- Section 7.0 – Reporting Requirements: details the frequency and content of the environmental monitoring reports submitted to the WDEQ/SHWD.
- Section 8.0 – References.

2.0 GROUNDWATER AND LEACHATE MONITORING PLAN

A total of 32 groundwater monitoring wells/piezometers are in the current groundwater monitoring network at the Landfill. In addition, 4 leachate wells (installed in the southern portion of the landfill at the refuse-soil interface), and 5 leachate sumps (installed in the northern trenches) are currently in the leachate monitoring program. Figure 1 shows the locations of each of the groundwater monitoring wells and piezometers, as well as the leachate wells and sumps. Hydrogeologic zones of completion are noted on Figure 1 where applicable.

Table 1 provides a summary of the construction history of monitoring wells and piezometers installed at the Landfill, dating back to 1975, including a list of associated reference documents. Construction details for each of the active monitoring wells and piezometers is included in Table 2 with the available associated boring logs/construction diagrams included in Attachment A. Each well and piezometer at the landfill was constructed with a protective surface casing and locked to prevent tampering.

2.1 PHASES OF GROUNDWATER MONITORING

Groundwater monitoring is divided into two phases at the Landfill: detection and assessment monitoring, following the general guidelines established in Chapter 2 of the WDEQ/SHWD Rules and Regulations (WDEQ 2017). Routine groundwater monitoring including gauging and sampling activities will be conducted in April and August of each year. Tables 3 and 4 present summaries of the sampling and fluid level monitoring plans for this landfill. Tables 5, 6, and 7 show the Solid Waste Rules and Regulations (SWRR) Chapter 2 Appendix A, Appendix B, and Appendix C Constituents, as required for detection and/or assessment monitoring.

2.1.1 DETECTION MONITORING

Detection monitoring includes the collection and analysis of groundwater samples where statistically significant impacts to groundwater have not been identified. Detection monitoring requires the analysis of a select list of constituents, followed by statistical analysis of the results to determine if there are increasing contaminant concentrations above background or baseline water quality. Groundwater samples collected during detection monitoring are analyzed for WDEQ/SHWD Appendix A (Table 5) and Appendix C (Table 7) constituents.

For routine detection monitoring, samples will be analyzed for Appendix A constituents on an annual basis and for Appendix C constituents on a semi-annual basis. However, for new monitoring well installations (see Section 2.4), baseline data will be developed by collecting samples quarterly for the first two years. Samples will be analyzed for

both Appendix A and C constituents at each sampling event. After the first two years, the sampling schedule will return to routine detection monitoring as described above.

2.1.2 ASSESSMENT MONITORING

Assessment monitoring is implemented whenever a statistically significant increase over background water quality has been detected for an Appendix A constituent, indicating potential impacts to groundwater quality. Assessment monitoring specifies a much broader range of analyses within selected wells to characterize the nature of groundwater impacts. Historically, a subset of wells developed in consultation with the WDEQ Administrator were sampled under assessment monitoring and analyzed for Appendix B (Table 6) constituents at least annually.

When an Appendix B or C constituent is detected at a statistically significant level above the applicable groundwater protection standard for that constituent, additional monitoring wells may be required to characterize the nature and extent of the release. The EE-series wells and P1E-9 are used to help define the nature and extent of VOC impacts first identified in compliance monitoring well SRM-6. Historically, samples collected from these wells were analyzed annually for the WDEQ/SHWD Appendix A constituent list plus the constituent dichlorodifluoromethane from the Appendix B list. Dichlorodifluoromethane is no longer sampled alongside Appendix A constituents; the current monitoring plan does not deviate from constituents listed in WDEQ/SHWD Appendices A, B, and C.

All wells scheduled for assessment monitoring (the subset of compliance wells described above) will be sampled semiannually for Appendix C parameters and annually for Appendix B (VOC). This group of wells also includes up-gradient wells in the Perched Water Bearing Zone and the Principal Unconfined Aquifer for statistical comparison. The assessment monitoring wells will be sampled for a reduced list of Appendix B constituents (only VOCs) based on an extensive data record that showed no significant detections of SVOCs or pesticides, herbicides or PCBs. The down-gradient wells scheduled for assessment monitoring will be sampled once every five years for the full Appendix B list, to confirm the lack of detection of these constituents.

The City will collect and analyze samples for the Appendix B (VOC) constituent list during the April sampling event. However, if there is insufficient liquid to complete the analyses, a second attempt will be made during the August event. Table 3 shows which wells are scheduled for assessment monitoring and Table 6 presents the Appendix B constituent list.

2.2 GROUNDWATER BEARING ZONES, AQUIFERS AND LEACHATE MONITORING POINTS

Four (4) saturated units have been identified beneath the Landfill. These include two (2) Perched Water Bearing Zones, the Principal Unconfined Aquifer, and the Semi-Confined (formerly called Confined) Aquifer. The perched

zones and the aquifers are not considered to be hydraulically connected beneath the Landfill. These water bearing zones and aquifers and the associated monitoring networks are described briefly in the following sections. Leachate wells are screened across the refuse-soil interface and are utilized to measure and characterize leachate generated beneath the unlined areas of the landfill. Leachate sumps are also monitored to assess the volume and characteristics of leachate generated from the lined areas. Additional monitoring wells, sumps, and piezometers may be installed at the landfill as part of investigation and monitoring activities.

2.2.1 PERCHED 1 WATER BEARING ZONE

The Perched 1 Water Bearing Zone has been identified as the uppermost water-bearing unit beneath the landfill. Intermittent saturated conditions have been encountered in the Spring Creek alluvium along the southern portions of the landfill extending into the Ogallala Formation to the east of Well SRM-6. Depth to water in the Perched 1 Water Bearing Zone in the southern portions of the landfill ranges between six (6) and 45 feet below ground surface (ft-bgs). East of the landfill, the perched zone is encountered at depths between 55 and 75 ft-bgs, and is limited at depth by a sandy clay layer in the Ogallala Formation. This zone is of limited extent both horizontally and vertically. Flow through the Perched 1 Water Bearing Zone is likely the result of channelized subsurface flow such as a sandy braided stream depositional environment.

In order to limit the amount of water entering the landfill from the Spring Creek drainage, a cut-off wall was constructed near the southern boundary where the creek enters the landfill. In addition, Spring Creek was re-routed to the south of the landfill. As part of that project, a pump was installed in an up-slope well to dewater any saturated alluvium in front of the cut-off wall. These efforts appear to be successful with minimal amounts of water entering the landfill through leakage around the cut-off wall or vertical infiltration through the landfill cap. In fact, many of the wells/piezometers screened across the Perched 1 Water Bearing Zone only intermittently contain groundwater, with limited water level data often precluding delineation of potentiometric surface mapping.

The Perched 1 monitoring wells will be gauged in April and August, of each year (Table 4). Monitoring wells and piezometers screened in the Perched 1 Water Bearing Zone are sampled under either assessment or detection monitoring, as outlined below and in Table 3:

- **Detection Monitoring:** Piezometers P-1, P-3, P-4, P-5, P-8, P-10, P-11 and Well S-9 are scheduled to be sampled for the Appendix A and C constituent lists in April and for the Appendix C list in August of each year.
- **Assessment Monitoring:** Groundwater samples are collected from Well SRM-6, P1E-9, EE-04 and EE-07 and analyzed for the Appendix B (VOC) and Appendix C lists in April, and the Appendix C list in August of each year.

The Perched 1 Water Bearing Zone is generally referred to as the Perched Water Bearing Zone (without numbering) throughout the documentation post-2003, since wells in the Perched 2 Water Bearing Zone were abandoned in 2003 as described below.

2.2.2 PERCHED 2 WATER BEARING ZONE

A Perched 2 Water Bearing Zone was identified in two Wells EE-01 and EE-02 installed along the eastern edge of the landfill. Monitoring Wells EE-01 and EE-02 were plugged and abandoned on June 9, 2003, because refuse was scheduled to be placed in close proximity to these two wells due to completion of planned development of the waste disposal area approved by the WDEQ/SHWD. The monitoring wells were plugged and abandoned in accordance with the provisions of Chapter 26, Section 11 of the Water Quality Division Rules and Regulations. The concrete pads were destroyed and the protective casings removed. Exposed portions of the well casings were cut off below the ground surface. The wells were then sealed with a bentonite grout from the total depth of the well to ground surface. A permanent marking was set into a concrete surface monument indicating the well identification and date of abandonment. The abandonment notices filed with the Wyoming State Engineer's Office were included in the May through August Environmental Monitoring Report submitted to the WDEQ/SHWD on October 28, 2003.

2.2.3 PRINCIPAL UNCONFINED AQUIFER

The entire landfill is underlain by an unconfined aquifer approximately 150 -160 ft-bgs. Thickness in the Principal Unconfined Aquifer varies from five (5) to 50 feet; groundwater flows from the southwest to the northeast at a gradient of approximately 0.01 to 0.02 feet per foot.

There are twelve (12) monitoring wells screened across the Principal Unconfined Aquifer at the landfill. Fluid levels from each of the wells screened in the Principal Unconfined Aquifer will be gauged in April, and August each year (Table 4). These wells will be sampled following the detection and/or assessment phase of groundwater monitoring, as outlined below and in Table 3:

- **Detection Monitoring:** Wells PU-1, PU-2, PU-3, PU-4, PU-5, and P3-1, P3-2 and P3-3 are scheduled to be sampled for the Appendix A and C constituent lists in April and for the Appendix C list in August each year.
- **Assessment Monitoring:** Groundwater samples will be collected and analyzed from Wells SRM-4, SRM-5, SRM-7 and SRM-8 for the Appendix B (VOC) and Appendix C constituent lists in April, and the Appendix C list in August each year.

2.2.4 SEMI-CONFINED AQUIFER

In 2013, a series of groundwater monitoring wells (GMW series) was installed to help define the subsurface hydrogeology specifically in the landfill expansion area. WDEQ asked for confirmation as to whether the GMW-series wells are confined and if they monitor the same water bearing zone as the C-series wells, a set of wells that had been described as monitoring a confined zone within the Ogallala Aquifer identified when the wells were drilled in 1999. The confined zone was said to be primarily composed of sand with minor amounts of clay, silt, and gravel which lies below the Principal Unconfined Aquifer unit at approximately 170 ft-bgs.

A transducer study was then completed to shed more light on the local subsurface hydrology as a condition of the WDEQ Location Standards Variance Approval. The results of the study indicated that the Confined Aquifer would be more appropriately called a Semi-Confined aquifer. The wells monitor different intervals but may represent a series of “stacked” water bearing zones that locally are connected to varying degrees. Some may meet the definition of confined (C-series) because water levels in certain wells are consistently above the screened zone, but the wells as a group do not appear to represent confined aquifer conditions. The Ogallala Aquifer in this part of the country is very complex and the degree of interconnectivity is difficult to accurately assess. It was recommended that the GMW and C series wells be plotted together to represent groundwater flow conditions below the Principal Unconfined Aquifer set of wells. USGS mapping shows a general northeasterly groundwater flow gradient that begins to change to the northwest consistent with what is also observed in the potentiometric surface maps constructed based on the C-series and GMW-series wells.

Approximately ten (10) municipal water wells operated by the Cheyenne Board of Public Utilities (BOPU) are located within a one-mile radius of the landfill site. These wells are installed at depths between 200 and 400 ft-bgs. Pumping from the wells operated by the Board of Public Utilities is intermittent as necessary to augment the City of Cheyenne’s water supply. Each of the water supply wells has the capacity to produce between 150 to 600 gallons per minute from the Ogallala Aquifer. The transducer study results indicated that even at high late-summer pumping rates, the BOPU pumping wells did not alter the aquifer flow direction from the regional flow pattern. The Cheyenne Board of Public Utilities intermittently collects groundwater samples from these wells and analyzes the samples for the Appendix A constituent list. This data is provided to the City and shared with WDEQ through the annual report. Three (3) C-series and 5 GMW-series wells will be gauged and sampled in April and August of each year according to the detection monitoring program (Tables 3 and 4).

2.2.5 LEACHATE WELLS

There are four active leachate wells screened into the refuse across the southern portions of the Landfill. Figure 1 shows the locations of each of the leachate wells. Historically, water had been sporadically present in over 11 leachate wells, based on fluid gauging results in April and August of each year. In the past, grab samples had been collected annually from Leachate Wells P1E-1, S-7A, and S-4 and analyzed for the Appendix A constituent list in order to quantify constituent concentrations detected in the leachate along the southern portion of the landfill. Constituent concentrations are primarily for information purposes to characterize the leachate quality and are not related to any compliance criteria. The City has been collecting leachate data for many years and additional sampling was not likely to provide added value. As a result, the four active leachate wells (P1E-1, S-1, S-4 and S-7A) are not being sampled but fluid levels are gauged during each sampling event (Table 4).

2.2.6 LEACHATE SUMPS

There are five (5) sumps installed to monitor and evacuate leachate generated beneath the northern trenches at the landfill. Figure 1 shows the locations of each of the leachate sumps installed at the landfill. Fluid levels in sumps NT-1 and NT-2 are manually gauged semi-annually in concert with the routine sampling events to the nearest 0.1-foot relative to the top of each sump. NT-1 and NT-2 are not pumped unless leachate depths approach 12-inches of accumulation above the liner. Fluid levels in these two sumps have been either dry or declining such that future pumping is unlikely.

Fluid levels in NT-3 through NT-5 are recorded using pressure transducers installed in each sump. Landfill staff record fluid levels and leachate pumping volumes read from the electronic display on a monthly basis. The pressure transducers in Sumps NT-3 through NT-5 will be verified every two years. The pressure transducers will be verified to the nearest 0.1-foot utilizing a water level indicator. If the measurement between the water level indicator and the pressure transducer differ by greater than 0.3 foot, the transducer will be recalibrated. The sumps are evacuated using dedicated submersible pumps programmed to prevent more than 12-inches of leachate from accumulating above the cell liner. Grab samples will be collected annually from the NT-5 leachate sump (if sufficient volume is present) and analyzed for the Appendix B (VOC) and C constituent lists in order to quantify constituent concentrations detected in the leachate (Table 3). Sump NT-5 was selected because it collects leachate from the most recently constructed cell and will produce leachate for the longest future time period. In addition, sample collection can be accomplished easily because of a sample port location on the discharge line and no confined space entry is required. Again, constituent concentrations are primarily for information purposes to characterize the leachate quality and are not related to any compliance criteria. As new landfill cells are constructed, the City proposes to sample one leachate collection area for the new cells for Appendix A and C constituents annually.

2.3 GROUNDWATER MONITORING PROCEDURES

This section details the procedures and techniques that will be followed for groundwater monitoring at the Landfill. Due to the heterogeneity of the hydrogeologic units encountered beneath the landfill, it may be necessary to modify these procedures as sampling progresses to ensure that representative groundwater samples are collected and submitted to the laboratory for analysis.

2.3.1 FLUID LEVEL MONITORING

Upon arriving at a well location, sampling personnel will check the well for damage and remove the lock and well cap. Any damage to the wellhead or casing, or any items requiring maintenance will be noted in the field sampling record and reported to the project manager so that the necessary repairs can be scheduled. Field personnel will wear new, sterile gloves during all preparation and sampling activities.

Fluid levels within each water bearing zone will be measured within a 24-hour period during the two scheduled sampling events (April and August). Fluid level measurements will be performed using a water level indicator accurate to 0.01-feet. The fluid level probe will be lowered down the center of the well casing. When the probe encounters water, an audio alarm is triggered. The measurement will be made from the pre-marked (surveyed) measuring point on the north side of the well casing. Manufacturer's instructions will be followed to ensure proper care of the fluid level probe. The total depth of the well will be measured annually to determine if the well is silting in and needs to be rehabilitated or replaced.

If non-aqueous phase liquid (NAPL) is observed in any well, the thickness will be measured with an interface probe. A typical interface probe gives an audible beeping sound when submerged in water and a solid tone when submerged in NAPL. The fluid level measurements will be recorded on the fluid level summary form (or similar) included in Attachment B.

All measurements during a particular monitoring event are made to the reference points with the same instrument, to ensure internal consistency. The vertical reference points (i.e. north side top of casing) are established by a licensed surveyor or under supervision of a professional engineer in relation to the nearest permanent benchmark. The horizontal well coordinates are established by marking distances in reference to the project site plan in triangulation or by surveying in reference to the nearest permanent benchmark. The exposed portion of the tape and the probe will be decontaminated before performing measurements at each monitoring well.

2.3.2 WELL EVACUATION PROCEDURES

Depending on the depth to water and the yield of a well, a dedicated pneumatic pump with dedicated tubing or a disposable bailer will be employed to purge groundwater from selected wells and piezometers. Field personnel shall note the approximate evacuation rate on the groundwater sampling forms (Attachment B).

Conventional sampling techniques will be used to evacuate and sample monitoring wells screened in the Perched Water Bearing Zone, as well as the leachate wells and sumps. The column of standing fluid in each well (casing volume) will be calculated based on the measured fluid level, total well depth, and well diameter. Where possible, the calculated well casing volume will be evacuated from the well a minimum of three times. Some wells may be pumped or bailed “dry” before three casing volumes are evacuated. If this occurs, groundwater samples will be collected following sufficient recharge to the well.

2.3.2.1 WELL EVACUATION USING A DEDICATED PUMP

There are currently eight wells with dedicated pumps at this facility. This includes pumps at the Semi-Confined Aquifer Wells C-2 and C-3 as well as pumps at the Principal Unconfined Aquifer Wells PU-1, PU-2, SRM-4, SRM-5, SRM-6, and SRM-8. These pumps are operated at speeds varying from 0.19 gpm to 0.29 gpm during evacuation and then allowed to pump a minimum of three well casing volumes. This is done by using an air compressor and discharge controller in collaboration with the dedicated pumps. Wells that have had low rechargeability during previous events will be purged to the extent possible; however, enough volume will be left in the well to ensure sufficient water is available to fill the laboratory required volume for analyses. The only well likely affected by this scenario is SRM-5 and will be determined through the volume of water in the well and recharge rate during evacuation.

2.3.2.2 GROUNDWATER EVACUATION USING BAILERS

Some wells will be evacuated using bailers due to the minimal volume of groundwater in the well. When collecting groundwater using a bailer, the bailer will be slowly lowered using twine or cable into the well to minimize agitation of the water table. The bailer will be raised to the surface manually or using an electric or battery operated auto bailer downrigger system. Wells that require an auto bailer downrigger system are C-1, GMW-03, GMW-04, GMW-06, and GMW-07. Disposable polyethylene bailers will be used to purge wells. Wells P3-1, P3-3, PU-3, and P1E-1 have had low rechargeability during previous events and will be bailed to the extent possible; however, enough volume will be left in the well to ensure sufficient water is available to fill the laboratory required volume for analyses.

2.3.2.3 ALTERNATIVE SAMPLING TECHNIQUES

In the future, the City of Cheyenne may opt to perform the sampling at the landfill using alternative sampling techniques. At this time, no decision has been made as to the alternative method or methods that will be used. If the City of Cheyenne decides to use an alternative sampling method in the future, detailed sampling methods will be submitted to the WDEQ/SHWD prior to any changes in sampling techniques at the landfill. The following is one alternative sampling techniques that may be considered by the City of Cheyenne:

- A low-flow “minimal drawdown” sampling methodology to evacuate groundwater and sample wells screened in the Principal Unconfined and the Confined Aquifers. Low-flow sampling involves removing water at the same rate as it flows through the formation minimizing the entrainment of sediment. Low-flow sampling will conform to USEPA recommended procedures outlined in the document, *Low-Flow (Minimal Drawdown) Groundwater Sampling Procedures* (Puls and Barcelona, 1995). The pumps would be installed so that the intake height is located in the interval of interest as determined in conjunction with the City of Cheyenne and the WDEQ/SHWD.

2.3.3 FIELD ANALYSES

Field parameters including specific conductance, pH, and temperature will be measured during groundwater evacuation and at the completion of sample collection. Field parameters will be recorded on the groundwater sampling form for each well. Physical characteristics of the groundwater will be noted and recorded during sampling (e.g. sediment, hydrocarbon sheen, color, odor, etc.).

2.3.4 SAMPLE WITHDRAWAL

Groundwater samples will be collected from wells using dedicated pump/tubing or from disposable bailers. In accordance with SWRR Chapter 2, Section 9.b.i.C.II, no samples will be field filtered prior to laboratory analysis. Sample agitation will be minimized during sampling and groundwater will be transferred directly from the sampling device into the sample containers, intermediary containers will not be used for sample collection. Sample bottles shall be filled as appropriate to minimize headspace. Volatile organic analysis (VOA) sample vials will be filled with zero-headspace. Field personnel will wear disposable gloves to prevent contamination of samples and equipment.

Groundwater samples will be submitted for analysis in accordance with the WDEQ/SHWD Appendix A list (Table 5), which includes VOCs by USEPA Method 8260B and total metals via USEPA Method 6020 or the Appendix B list (Table 6) which includes additional VOCs and total metals, as well as, semi-volatile organic compounds (SVOCs) by USEPA Method 8270C; and pesticides, herbicides/fungicides, and polychlorinated biphenyls (PCBs) via USEPA Methods 8081A, 8151A, and 8082, respectively. In addition, the City of Cheyenne will monitor the Appendix C list (Table 7) of geochemical parameters from each of the wells included in the sampling network.

Sample bottles will be preserved in the laboratory before being sent to the site. The samples will be collected and analyzed in general accordance with the Test Methods for Evaluating Solid Waste, USEPA, SW-846 (EPA 2007). The collection order for water quality samples is:

1. VOCs
2. SVOCs
3. Extractable organic compounds
4. Total metals
5. Inorganic compounds and nutrients

A trip blank prepared by the laboratory will be shipped with the groundwater samples. The samples will be labeled and placed in an opaque cooler and stored on ice after collection. Glass containers shall be protected against breakage during transport to the laboratory. A chain-of-custody form will accompany the samples to the laboratory. The samples will be submitted to TestAmerica in Denver, Colorado or an equivalent laboratory.

2.3.5 FIELD MEASUREMENTS, CALIBRATION, AND DECONTAMINATION

Field instruments will be used to analyze air and/or water samples during monitoring activities at the landfill. Field measurements provide information critical to assessing health and safety issues, evaluating the nature, degree, and extent of impacts, and ensuring representative samples are selected for laboratory analyses. Therefore, calibration and maintenance of field instruments is important to achieving the overall project objectives.

Numerous types and brands of field instruments are currently available for use during environmental assessment projects. Maintenance and calibration procedures vary depending on the type, manufacturer, model, and age of the instruments. Therefore, specifying calibration and maintenance procedures for one or many types of instruments is not considered to be practicable in the SLEMP.

Instruments and equipment used to gather, generate, or measure environmental data will be calibrated with sufficient frequency and in a manner that is consistent with the manufacturers' specifications. A record of calibration will be maintained documenting calibration results and will include the date, standards, personnel, and calibration results.

Sampling equipment and containers will be decontaminated prior to sampling, either in the laboratory and/or in the field. The sampling devices will be decontaminated in the field immediately prior to and after sampling each site. In general, decontamination will consist of a detergent/water wash, tap water rinse, and distilled water final rinse followed



by drying before subsequent use or storage. Decontaminated sampling devices and containers will be stored in contaminant free locations or containers until use. Glass jars, etc. may be stored until use in clean sample coolers.

2.3.6 DOCUMENTATION REQUIREMENTS

All field activities and samples collected will be documented in order to verify compliance with the SLEMP. Field forms will be used to document the monitoring activities. Information generated from sampling activities will be documented on the appropriate forms including:

- Fluid Level Summary
- Groundwater Sampling Form
- Record of Calibration
- Chain-of-Custody

Incoming documents will be dated and filed. If distribution is necessary, the documents will be distributed to project personnel. Examples of all of these forms are included in Attachment B.

2.3.7 CHAIN-OF-CUSTODY

A sample is under custody if it is in the field personnel's possession or if the sample was in the field personnel's possession and then locked up, or placed in a designated secure area. The following procedures will be used in the field:

1. The field personnel will be responsible for the care and custody of the samples collected until they are delivered to the analytical laboratory or entrusted carrier. As few people as possible will handle the samples.
2. Field forms will be used to document the data collection activities such as the date, start and end time of activities, field personnel name, weather conditions, sample location, equipment used to collect sample, depth of sample, time of collection, sample description, sample identification number, and the volume and number of containers. Entries will be made in ink with no erasures. A single stroke will be used to cross out incorrect information; corrections will be dated and initialed.
3. A chain-of-custody will be completed in the field to the fullest extent possible prior to sample shipment. It will include the following information: sample number, time collected, source of sample and location, depth from which sample was taken, preservative, and name of field personnel. The chain-of-custody will be filled out in a legible manner, using ink, and signed by field personnel. Similar information will be provided on the sample label,

which will be securely attached to each sample bottle/container. In addition, sampling forms will be used to document collection, filtration (if required), and preparation procedures.

2.3.8 CUSTODY FOR SHIPMENT

The following procedures will be used when transferring custody of samples:

1. Samples will always be accompanied by a chain-of-custody completed according to the protocols outlined in this section. When transferring samples, the individuals relinquishing and receiving them will sign, indicate the date, and note the time on the chain-of-custody. Upon arrival at the laboratory, internal custody procedures will be followed, in accordance with standard laboratory procedures.
2. Samples will be packaged properly for shipment and dispatched to the laboratory for analysis with individual custody records accompanying each shipment. Shipping containers will be sealed for shipment to the laboratory. The method of shipment, courier name, and other pertinent information will be entered in the remarks section of the chain-of-custody.
3. All shipments will be accompanied by the chain-of-custody form identifying the contents. The original record will accompany the shipment, and a copy will be retained by the field sampler.
4. Proper documentation will be maintained for shipments by the common carrier.

2.3.9 SAMPLE SHIPMENT PROCEDURES

Sample containers will be packed securely in a cooler to minimize the potential for breakage during shipment. Packing materials, such as poly-foam or bubble wrap, shall be used to protect sample containers in the cooler.

The following procedures will be followed when shipping samples for laboratory analysis:

1. Samples requiring refrigeration will be promptly chilled with ice to a temperature of four (4) °C and packed in an insulated cooler for transport to the laboratory.
2. Only shipping containers that meet all applicable state and federal standards for safe shipment will be used for this project.
3. To provide a means for detecting any potential tampering during shipment, custody seals will be affixed to cooler and tape wrapped entirely around the cooler.

4. The chain-of-custody will be placed inside the shipping container in a sealed plastic envelope.
5. Shipment will be made by overnight courier or picked up by laboratory personnel within 24 hours of sampling or as soon as practicable.

2.3.10 SYSTEM AUDITS

Performance and system audits of both field and laboratory activities will be performed on a periodic basis, as appropriate, to assure that the sampling and analysis are performed in accordance with the procedures established in the SLEMP. Examples of audits that may be performed are as follows:

- Field inspections to confirm that the groundwater monitoring program was conducted correctly; field measurements were made accurately; equipment was thoroughly decontaminated; samples were collected and handled properly; and field work was accurately documented.
- A comprehensive review of the analytical report submitted by the laboratory to verify all requested analyses were performed; sample holding times were met; the data was generated using the appropriate methodology, the level of QC effort and reporting; and the analytical results were in conformance with the prescribed acceptance criteria.
- Oversight of the field and data validation activities to ensure data management proceeds in an organized and expeditious manner.

2.3.11 CORRECTIVE MEASURES

Any person identifying an unacceptable condition will notify the project manager. The project manager will be responsible for developing and initiating appropriate corrective action and verifying that the corrective action has been effective. Corrective actions may include repeating measurements, re-sampling and/or re-analyzing samples, and amending or adjusting sampling procedures. If warranted by the severity of the problem (e.g., if monitoring wells require re-sampling or if the project schedule may be affected), the WDEQ/SHWD will be notified.

2.4 FUTURE EXPANSION AREA WELLS

Variance condition #1 requires the installation of additional groundwater monitoring wells downgradient of the expansion area at a spacing not to exceed 500 feet. Four additional monitoring wells will be installed prior to waste placement in the expansion area to meet this requirement. Well locations will be provided to WDEQ prior to installation.

Eight quarterly sampling events for Appendix A and C constituents will be conducted to establish baseline conditions for the new expansion area wells. At a minimum, four of the eight quarters of baseline monitoring will occur prior to waste placement. The sampling frequency will be reduced to semiannual thereafter to be consistent with the detection monitoring sampling schedule, as described in Section 2.1.1 above.

2.5 OFF-SITE WATER WELL MONITORING PROGRAM

Variance Conditions # 9 and #10 specify that the City of Cheyenne implement a groundwater monitoring program for water wells permitted or certificated for domestic or stock use located within ½ mile of the expansion area facility boundary, provided that the City is given access to the specified water wells by the property owner.

2.5.1 BASELINE MONITORING PROGRAM

In accordance with Variance Condition #10, the City will implement a baseline water quality monitoring program for off-site water wells. Following lifetime permit approval, the City will contact the permit holders listed in Table 8 via certified mail, taking care to check for any new wells installed since the initial tabulation. In the event landowners do not respond to the first certified mailing, a second mailing will be sent to landowners seeking permission. Landowners will be asked to provide written permission to allow the City to inspect water well integrity and take water quality samples for Appendix A and C constituents for four quarterly baseline sampling events and potential future expanded monitoring events. The City will maintain records of the mailings and signed approvals for each property contacted.

Every effort will be made to take samples from the subject wells upstream of any treatment system that may have been installed. Sampling procedures for each property will be on a location by location basis and dependent upon the conditions and use of the property well. Therefore, once initial inspection is completed at each location, sampling protocol will be determined and submitted to WDEQ prior to initiating the first sampling event. Field technicians will document sampling locations and any modified procedures in the associated field notes. Following completion of sampling, monitoring data will be submitted to the WDEQ.

Note that Table 8 does show a well owned by the City and a well owned by BOPU. Consent for locating the landfill in proximity to these wells was previously provided by these two entities. While the City and BOPU signed letters of consent for the location standards in the expansion area, this does not mean that the City and BOPU also waived the additional sampling on their permitted wells. These two entities will be included as part of the baseline monitoring program and will be given the opportunity to decline and/or accept this monitoring. In addition, water quality analyses is routinely completed for the BOPU well (Koppes #6) and is already provided to WDEQ as part of the Landfill annual environmental monitoring report. Please also note that the King Ranch well, while listed as being within ½ mile of the

expansion area, is actually closer to the existing landfill and not located hydraulically down-gradient of the expansion area. After consultation with WDEQ, it was not determined to be necessary to include the King Ranch well in the off-site monitoring program.

2.5.2 EXPANDED MONITORING PROGRAM

In accordance with Variance Condition #9, if constituents in any groundwater monitoring well located downgradient of the expansion area exceed the EPA Maximum Contaminant Level, the City will institute a monitoring program for the off-site water wells within 90 days, and annually thereafter, for Appendix A, B and C constituents. The variance language states Appendix B and baseline constituents, however, given the January 3, 2017 revisions to Chapter 2 of the SWRR, the Appendix A, B and C constituent lists will be used.

3.0 GROUNDWATER DATA QUALITY ASSURANCE/QUALITY-CONTROL

The overall quality assurance objectives for the environmental monitoring at the Happy Jack Landfill is to provide quantitative and qualitative measures to produce legally defensible data used to evaluate subsurface quality beneath the landfill. The objectives of the quality assurance/quality control (QA/QC) program are to:

- Verify that all procedures are documented, including any changes from the protocols established in the SLEMP.
- Verify that all sampling and analytical procedures are conducted according to sound scientific principles.
- Monitor the performance of the field sampling team and laboratory with a systematic audit program and provide corrective actions necessary to assure quality.
- Evaluate the quality of the analytical data through a system of quantitative and qualitative criteria.
- Verify that all data and observations are properly recorded and archived.

Previous sections of this report have documented the procedures for field sampling, chain-of-custody, and laboratory analysis in order to satisfy many of the aforementioned objectives. This section addresses the qualitative and quantitative criteria which will be used to evaluate the quality of the analytical data collected during field activities and laboratory analysis.

3.1 QUALITY CONTROL PARAMETERS

The level of quality control effort will generally be consistent with the United States Environmental Protection Agency (USEPA) Contract Laboratory Program (CLP) National Functional Guidelines (USEPA 2014a, 2014b, 2013, 2004, and 1999). The data quality elements that will be evaluated include:

1. **Blind Duplicate Samples:** blind duplicate (field replicate) samples will be collected to evaluate precision associated with the reproducibility of sampling techniques and the homogeneity of sample matrices. Replicate samples will be collected for each matrix at a frequency of ten percent (10%) or one (1) per every ten (10) samples collected and submitted for laboratory analysis.

If less than ten (10) samples are collected during a particular sampling event, one blind duplicate sample will be collected. Since the replicate will be “blind” to the laboratory, it will have a coded identity on its label and on the chain-of-custody record form. The actual sampling location and identification will be recorded on the sampling forms.

2. **Trip Blanks**: one (1) trip blank will be submitted for VOC analysis with each cooler containing aqueous sample requiring VOC analyses. Trip blanks are supplied by the laboratory with the sampling containers at the start of field activities and accompany the sample containers throughout the project.
3. **Equipment Blanks**: equipment blanks will be prepared and submitted for laboratory analysis to determine if cross-contamination has occurred during sampling, and to verify that equipment decontamination procedures are effective. Equipment blanks are submitted for analysis of the same constituent list required for the associated field samples. Equipment blanks will be preserved in the same manner as the associated samples. Equipment blanks will be shipped with the field sample containers. Equipment blanks will be prepared at a minimum of one per sampling event if required.
4. **Method Blanks**: method blanks will be prepared and analyzed by the laboratory at a frequency of one per every 20 samples or one per sample batch. The method blank is a “clean” matrix similar sample prepared and analyzed by the laboratory. Analysis of the method blank is used to identify laboratory derived contaminants introduced during sample preparation, extraction, and analysis.
5. **Matrix Spike/Matrix Spike Duplicates (MS/MSD)**: matrix spike/matrix spike duplicates will be prepared and analyzed by the laboratory at a frequency of five percent or one per every 20 samples submitted to the laboratory. MS/MSDs are samples in which known spike concentrations of analytes are added to the collected sample prior to extraction and analyses. The recoveries for spiked compounds can be used to assess how well the method for analysis recovers target compounds within the submitted matrix. Evaluation of the MS/MSD is used to evaluate laboratory accuracy and precision, as well as to determine any potential matrix related interferences within the analytical results.
6. **Laboratory Control Samples (Performance Evaluation Samples)**: laboratory control samples (LCS) are “clean” matrix similar samples prepared by the laboratory and spiked with a known concentration of constituents prior to extraction and analysis. The LCS is used to evaluate laboratory accuracy and method compliance. The LCS will be prepared and analyzed by the laboratory at a frequency of one (1) per every 20 samples analyzed or one per sample batch.
7. **Surrogate Spiking**: surrogate compounds consist of laboratory derived compounds that are introduced to each sample submitted to and prepared by the laboratory prior to extraction and analysis. The surrogate compounds spiked to each sample are specific to the laboratory analytical method. Quantification of the surrogate compounds allows for determination of matrix effects and laboratory performance on individual samples.

3.2 QUALITY ASSURANCE AND QUALITY CONTROL MEASURES

Four QA/QC measures will be evaluated for samples collected as part of routine monitoring and assessment activities:



- Precision
- Accuracy
- Method compliance
- Completeness

A discussion of each quantitative and qualitative measure is outlined below.

3.2.1 PRECISION

Precision is the measure of variability of sample measurements. Measurements of data precision are necessary to demonstrate the reproducibility of the analytical data. Field precision is determined by a comparison of field duplicate sample results. Laboratory precision is determined by examining the laboratory duplicate results. Evaluation of both the field and laboratory duplicates for precision is accomplished using the relative percent difference (RPD). The RPD is defined as the difference between the primary and duplicate samples divided by the mean and expressed as a percentage. Field duplicate RPD limits are set at 0-30% and laboratory RPD limits reference published or laboratory control charted limits.

3.2.2 ACCURACY

Accuracy is the relationship of the reported data to the accepted or “true” value. Accuracy is a measure of sampling and analysis bias. Biased results generally arise from personnel, instrument, or analytical method influences. The accuracy of the methods used for the analyses of the samples will be evaluated through the analysis of calibration standards, MS/MSD recoveries, surrogate spike recoveries, and blank results. Analytical accuracy is expressed as the percent recovery of a spiked constituent reported in the surrogate, LCS, or MS/MSD; and is expressed as the concentration of the primary sample (zero in the case of the LCS) subtracted from the total concentration divided by the spiked concentration.

$$\text{Accuracy} = (C^T - C^P) / C^S$$

Where: C^P = concentration of primary sample

C^T = total reported concentration

C^S = concentration of spike constituent

The spiked concentration is specified by the laboratory quality control requirements. The compounds used and their accuracy limits are specified in the respective methods listed in SW-846.

3.2.3 METHOD COMPLIANCE

Method compliance is a qualitative evaluation of the field and laboratory procedures used to collect routine monitoring data against the prescribed methods outlined in the SLEMP, SW-846, and the laboratory quality assurance project plan. Laboratory method compliance is determined by reviewing the holding times, detection limits, surrogate recoveries, methods blanks, and laboratory control samples against method specific requirements. Method compliance for sample collection is evaluated by reviewing sampling forms, chain-of-custody, and field notes.

3.2.4 COMPLETENESS

Completeness is a measure of the percentage of analytical data that meet all the acceptance criteria for precision, accuracy, and method compliance. Completeness is the overall ratio of the number of samples planned versus the number of samples with valid analyses.

The laboratory shall provide data that meet QC acceptance criteria for 90% or more of the requested determinations, which is standard industry practice. The ability to meet or exceed a completeness objective is dependent on the nature of samples submitted for analysis. Completeness goals can still be achieved if the data requires qualification as estimated above or below laboratory reporting limits (i.e. data of known quality even if not flawless, can be suitable for specific project goals). If the percent completion limits cannot be met because data is qualified as rejected or non-useable for the project objectives, the laboratory or environmental compliance contractor may be responsible for re-sampling and/or re-analysis dependent on the nature of the errors.

4.0 GROUNDWATER DATA STATISTICAL ANALYSES

Annually, statistical analysis will be completed on the analytical results collected over the year following receipt and validation of the analytical data. Multiple statistical tests will be performed to determine if there is statistically significant evidence of impacts to any of the aquifers/groundwater bearing zones. Groundwater monitoring results dating back to September 1982 will be included in the statistical analyses. Statistical analyses will be completed using selected up-gradient (background) and down-gradient (compliance) wells screened in the Semi-Confined Aquifer, Principal Unconfined Aquifer, and Perched Water Bearing Zones.

The analytical procedures used to evaluate groundwater quality beneath the landfill meet and exceed the requirements stipulated in SWRR Chapter 2, Section 9.b.i.C.VI (WDEQ 2017). The statistical procedures include: determining the distribution of the data, calculating basic summary statistics, interwell prediction interval testing, intrawell trend analysis and compliance limit comparisons to groundwater protection standards. Statistical analysis is not required when all concentrations of constituents are reported below the practical quantitation limit (PQL), aka reporting limit (RL). Statistical analysis will be completed in general accordance with EPA guidance (EPA 2009).

4.1 GENERAL STATISTICS

Prior to conducting the statistical analysis, the distribution of the data for each constituent will be determined. The Shapiro-Wilk test for normality is the default test and is used when the sample size is less than 50. There are often less than 50 data points available for groundwater analytical results for each of the wells included within the monitoring well network. Once the sample size exceeds 50, the Shapiro-Francia test for normality will be used.

After the distribution of the data is determined for each constituent, the data set will be examined for possible outliers. In accordance with the Statistical Analysis for Groundwater at Resource Conservation and Recovery Act (RCRA) Facilities, Unified Guidance (USEPA 2009) observed outliers will not be excluded from the data set unless it can be demonstrated that the outlier results from a data entry or laboratory error.

Summary statistics, to include the mean, median, standard deviation, and variance will then be calculated for each of the constituents detected in groundwater collected from each monitoring well.

4.1.1 NON-DETECT DATA

Laboratory results reported as non-detect (ND) will be entered into the statistical analysis as a value of one-half the PQL. Estimated values reported below the PQL but above the method detection limit (MDL) will be treated as valid measurements for statistical purposes.

4.2 INTERWELL STATISTICS

Interwell prediction limits (or intervals) will be calculated to identify constituents with statistically significant differences in concentrations between up-gradient and down-gradient wells. Upper prediction limits will be constructed to contain, with $(1-\alpha)$ probability ($\alpha = 0.05$), the next measured concentration. The interwell procedure calculates an upper prediction limit using the background wells and then compares the most recent concentration reported in each compliance well to the upper prediction limit.

Table 9 provides the compliance and background monitoring wells utilized for interwell analysis in each of the aquifers/water bearing zones. The upgradient/background wells were chosen in consultation with WDEQ staff based on joint review of historical data. Background wells were chosen that represent the same aquifer/water-bearing unit as the compliance wells in each case.

In the Perched Water Bearing Zone, wells P-5 and S-9 were chosen as background wells as they appeared to be located upgradient of landfill activities, however, background conditions in a perched water zone may be difficult to determine as hydraulic conditions are not well defined. Please note that some of the Perched Water Bearing Zone wells have been dry for an extended period of time and will not always be included in the statistical analysis. This is sometimes the case for wells EE-04, EE-07, S-9 and P1E-9. Deviations from the standard analysis will be noted in future reports.

4.3 INTRAWELL STATISTICS

Intrawell compliance tolerance interval analyses will be performed for down-gradient wells for Appendix A and B constituents which exceed the interwell prediction limits. This approach focuses intrawell compliance statistical analyses on those down-gradient wells and constituents with a statistically significant increase over background water quality.

The Mann-Kendall/Sen's Slope trend analysis will be used to determine if there is a significantly increasing trend in constituent concentrations reported in groundwater samples collected from any of the wells included in the monitoring network. An increasing slope is indicated by a positive Mann-Kendall statistic and an estimated Sen's Slope greater than zero. Time series plots and the data will be examined to ensure that a significant slope is not an artifact of changes

in detection limits over time. The trend analysis will not be used for compliance evaluations, but may provide important information regarding changes in groundwater chemistry, both up-gradient and down-gradient.

4.4 COMPARISONS TO GROUNDWATER PROTECTION STANDARDS

A 95% upper tolerance limit will be calculated for constituents detected in groundwater samples collected from monitoring wells that exhibit significantly higher concentrations compared to those reported in background wells (interwell analysis). The calculated 95% upper tolerance limit will be compared to the appropriate Groundwater Protection Standard (GWPS).

4.4.1 TOLERANCE LIMIT ANALYSIS

Tolerance limit analysis is a component of compliance limit testing. SWRR Chapter 2 Section 9.b.i.C.VI.3 (WDEQ 2017) recommends the use of tolerance limits or prediction intervals as a method to compare background wells to compliance wells. The 95% upper tolerance limit is currently being used to compare constituent concentrations measured in compliance wells to groundwater protection standards. When variation in the data set is large, the upper tolerance limit is often estimated well above actual constituent concentrations, which is the case with many constituents reported in groundwater samples collected from monitoring wells installed at the City of Cheyenne Sanitary Landfill.

In addition, if the distribution of analytical data for a constituent reported within a monitoring well is not normally distributed, the upper tolerance limit is estimated as the highest or the second highest concentration detected for that constituent, whichever provides the best estimate of the 90th percentile to the dataset with 95% confidence. This approach allows some flexibility when the data do not conform to an identifiable distribution and standard mathematical models cannot be used. If the highest or second highest reported concentration is greater than the groundwater protection standard, then the tolerance limit will consistently predict an exceedance above the groundwater protection standard, even if the remaining results are non-detects at the reporting limit. Historical statistical analysis has shown that many of the constituent concentrations reported in monitoring wells at the City of Cheyenne Landfill are not normally distributed and therefore the tolerance limit analysis could produce positive results driven by a single historic extreme value, potentially falsely indicating an exceedance above the groundwater protection standard.

5.0 LANDFILL METHANE MONITORING PLAN

A methane venting system was installed in the filled areas at the south end of the landfill before placing the final cap in 1991. Quarterly monitoring of methane concentrations in these vents was conducted from March 1992 through September 1993. Methane concentrations over this period were reduced from greater than 1,000 parts per million (ppm) to less than 1.0 ppm. Methane monitoring was discontinued along the south end of the landfill in September 1993.

A methane monitoring well (MMW-1) was installed between the North Trench and the newly constructed equipment building located north of the disposal cells. Methane concentrations have been measured in Well MMW-1 since September 1996, with concentrations sporadically exceeding 25% of the LEL. At that time, action limits for the perimeter of the landfill were established at 25% of the LEL. In correspondence dated December 7, 1998, the WDEQ/SHWD stated that additional measures to address the methane detections in MMW-1 were not necessary. Methane concentrations at MMW-1 had been more recently reported exceeding 100% of the LEL. This monitoring point is not located at the landfill boundary and is not considered a compliance methane monitoring point. The City has scheduled to plug and abandon MMW-1 in 2017 in favor of proposed methane monitoring probes along the facility boundary.

5.1 METHANE MONITORING AT THE LANDFILL BOUNDARY

Refer to the Addendum to the Sanitary Landfill Environmental Monitoring Plan (SLEMP), Methane Gas Sampling and Analysis Plan, Happy Jack Landfill, prepared by Golder and Associates, Inc. (2017; Appendix H-2 in Lifetime Permit), for the monitoring procedures for proposed gas monitoring probes located along the facility boundary. Methane monitoring at the facility boundary will be conducted quarterly in February, April, August, and December each year.

5.2 METHANE MONITORING OF EQUIPMENT BUILDINGS

The equipment building is constructed on a slab on grade beneath the shop area and on grade in the equipment storage bays. The building has a flexible membrane liner buried beneath the earth floor and concrete slab. The building is not equipped with a methane venting system. However, the building is equipped with two wind propelled turbine ventilators over the garage area and exhaust fans over the garage and shop area. The shop area and recycle building will also be monitored. Overhead doors on the buildings being monitored will be closed for a minimum of one-hour before testing.

A portable gas meter capable of monitoring oxygen and methane concentrations and % LEL will be utilized. Before collecting readings, the meter will be allowed to warm up and calibrated using methane as the calibration standard. Ambient air concentrations shall be recorded over a five-minute interval. The meter will be placed at a height of approximately three feet above the ground and allowed to equilibrate. Methane, oxygen, and percent LEL will be recorded every minute over a five-minute interval. After completing monitoring in the equipment area, the technician will proceed to the shop area and recycle building and repeat the sampling in those locations. Ambient air concentrations will be recorded on the methane monitoring field forms or similar forms included in Attachment B.

6.0 NOTIFICATION REQUIREMENTS

This section outlines the notification requirements and follow up actions for:

- Methane concentration above 25% LEL in facility buildings or above 100% LEL at compliance points at the landfill boundary
- Groundwater detections above GWPS
- Statistically significant increases in contaminant concentrations in groundwater

6.1 METHANE MONITORING NOTIFICATION REQUIREMENTS

If a methane level exceeding 25% of the LEL is detected in the landfill equipment building, shop building and/or recycle building, the building should be evacuated immediately, electricity shut off, contact made with the City of Cheyenne Public Works Director's office, the City of Cheyenne Fire Department, and the WDEQ/SHWD. Within seven (7) days of the exceedance, a copy of the methane test data will be placed in the operating record, along with a written description of the steps taken to protect human health. Within 60 days of the exceedance, a remediation plan will be implemented, which has been approved by the WDEQ/SHWD, and a copy of that plan placed in the operating record.

Please refer to the "Addendum to the Sanitary Landfill Environmental Monitoring Plan (SLEMP), Methane Gas Sampling and Analysis Plan, Happy Jack Landfill", prepared by Golder and Associates, Inc. (2017; Appendix H-2 in Lifetime Permit), for an explanation of the notification requirements should methane levels exceed the LEL at the gas monitoring probes to be located along the facility boundary.

6.2 DETECTION MONITORING NOTIFICATION REQUIREMENTS

If the concentration of an Appendix A constituent in any downgradient well located in the Perched Water Bearing Zone, Principal Unconfined Aquifer, or Semi-Confined Aquifer represents a statistically significant increase over background water quality during an annual review of available data, the City of Cheyenne Public Work's Office will be immediately notified. The WDEQ/SHWD will be contacted and a written report placed in the facility operating record within 14 days. Assessment Monitoring will begin within 90 days or in an acceptable time frame approved by the Administrator. Alternatively, the City may demonstrate to the WDEQ/SHWD that the statistically significant water quality difference is not due to the solid waste disposal facility, but that the difference is due to another source of pollution, error in sampling, analysis, or statistical evaluation, or natural variation in groundwater quality. The City of Cheyenne shall prepare a report documenting this demonstration, and following approval by the WDEQ/SHWD place

the report in the operating record for the facility. If the report is approved, the City of Cheyenne shall continue Detection Monitoring. If, after 90 days, a successful demonstration is not made, the City of Cheyenne will initiate an Assessment Monitoring program as required in SWRR Chapter 2, Section 9 unless an alternate timeframe is approved by the Administrator.

The WDEQ/SHWD will also be notified in writing and a notice placed within the operating record (SWRR Chapter 2, Section 9.b.i.D) within 14 days if the concentration of an Appendix A or Appendix C constituent with an MCL or a class of use based limit in the Water Quality Rules and Regulations (WDEQ 2015) is found to be a statistically significant increase over background water quality. The City will request in writing that WDEQ/SHWD establish a groundwater protection standard (WDEQ 2009) for that constituent. Within 30 days of the establishment of that standard, the City will determine whether the concentrations represent a statistical exceedance of the groundwater protection standard and notify the WDEQ/SHWD in writing within 14 days should that be the case.

6.3 ASSESSMENT MONITORING NOTIFICATION REQUIREMENTS

If there is a detection of any Appendix B constituent as part of Assessment Monitoring, the City of Cheyenne Public Works Director's Office will be notified. The WDEQ/SHWD will be notified within 14 days and a notice will be placed in the operating record identifying the Appendix B constituents that have been detected.

If the concentrations of all Appendix B constituents are at or below background values using the approved statistical procedures for two (2) or more consecutive sampling events, the City of Cheyenne must notify the WDEQ/SHWD and may return to Detection Monitoring.

If one or more Appendix B constituents are detected at statistically significant levels above the GWPS during an annual review of available data, the City of Cheyenne Public Works Director's Office will be notified. Within 14 days of this finding, a written report will be placed in the operating record identifying the Appendix B constituents (SWRR Chapter 2, Section 9.b.i.E). The WDEQ/SHWD Administrator, and appropriate local government officials as determined by the Administrator, will be notified. The City is also required to notify all persons who own or reside on the land that directly overlies any part of the plume of contamination, if that plume has migrated off-site.

7.0 REPORTING REQUIREMENTS

This section details the frequency and content of the routine progress reports submitted to the WDEQ/SHWD.

7.1 EVENT REPORT

A summary event report will be submitted to WDEQ/SHWD for the April and August monitoring events through the operating life of the landfill and through the post-closure period. Paper copies of each event report will be submitted within 60 days of receipt of complete and accurate analytical results along with electronic data, in the format specified by the WDEQ/SHWD Administrator. The event report will include:

- List of monitoring wells included in the Groundwater Monitoring Program
- Appendices containing field data, laboratory analytical data, quality control data, and chain of custody forms
- Fluid level summary tables and electronic data
- Groundwater analytical data tables and electronic data
- Potentiometric surface maps for the Semi-Confined and Principal Unconfined Aquifers
- Gradient, flow direction and velocity of groundwater in the Semi-Confined and Principal Unconfined Aquifers
- Results of quarterly methane monitoring
- Results of off-site monitoring events, if applicable

7.2 ANNUAL REPORT

Paper copies of a comprehensive annual report will be submitted to the WDEQ/SHWD prior to April 1 of the following year, along with required electronic data in the format specified by the WDEQ/SHWD Administrator. The annual monitoring period is January 1 through December 31. The annual reports will include:

- Summary of field and laboratory work completed during the previous year
- Documentation of operating procedures and field methods utilized
- Analytical data summary tables including methane monitoring results at both the facility boundary and the facility structures in accordance with May 6, 2008 Memorandum (Groundwater Monitoring Data Reporting Format for Municipal Solid Waste Landfills) and Section 3.2 of the Methane Gas Sampling and Analysis Plan (Golder and Associates 2017; Appendix H-2 in Lifetime Permit)
- Annual monitoring leachate data from NT-5

- Description of statistical analyses employed
- A summary of results from the statistical analyses and significant statistical increases
- Data disk of analytical results
- Summary of off-site monitoring data, if applicable
- Conclusions regarding the potential impact of landfill activities upon each aquifer and water bearing zone
- Recommendations for modifications to the Environmental Monitoring Program

8.0 REFERENCES

- Golder Associates, Inc. 2017. Addendum to the Sanitary Landfill Environmental Plan (SLEMP), Methane Gas Sampling and Analysis Plan, Happy Jack Landfill, Cheyenne, Wyoming. Lakewood, Colorado.
- Puls, R. and Barcelona, M. 1995. Low-Flow (minimal drawdown) ground-water sampling procedures. Washington D.C.: U.S. Environmental Protection Agency, Office of Research and Development, Office of Solid Waste and Emergency Response. 12 p.
- United States Environmental Protection Agency (USEPA). 2014a. USEPA Contract Laboratory Program (CLP) National Functional Guidelines for Superfund Organic Methods Data Review. Document Number EPA-540-R-014-002 Office of Superfund Remediation and Technology Innovation, Washington, D.C. August 2014.
- USEPA. 2014b. USEPA CLP National Functional Guidelines for Inorganic Superfund Data Review. Document Number EPA-540-R-013-001, Office of Superfund Remediation and Technology Innovation, Washington, D.C. August 2014.
- USEPA. 2013. USEPA New England Environmental Data Review Supplement for Regional Data Review Elements and Superfund Specific Guidance/Procedures, EQADR-Supplement0, April 2013.
- USEPA. 2012. 2012 Edition of the Drinking Water Standards and Health Advisories. Office of Water, Washington, D.C. (EPA 822-S-12-001). April 2012.
- USEPA. 2009. Statistical Analysis of Groundwater Monitoring Data at RCRA Facilities, Unified Guidance. Office of Resource Conservation and Recovery, Washington, D.C. (EPA530-R-09-007).
- USEPA. 2007. Test Methods for Evaluating Solid Waste, SW-846, Revision G. Office of Solid Waste, Washington, D.C.
- USEPA. 2004. USEPA CLP National Functional Guidelines for Inorganic Data Review. Document Number EPA 540-R-04-004, Office of Superfund Remediation and Technology Innovation, Washington, D.C. October 2004.

USEPA. 1999. USEPA CLP National Functional Guidelines for Organic Data Review. Document Number EPA 540/R-99/008, Office of Superfund Remediation and Technology Innovation, Washington, D.C. October 1999.

Wyoming Department of Environmental Quality (WDEQ), Solid and Hazardous Waste Division. 2017. Solid Waste Rules & Regulations, Chapter 2, Sanitary Landfill Regulations. Cheyenne, WY. January 3.

Wyoming Department of Environmental Quality (WDEQ), Solid and Hazardous Waste Division. 2009. Memorandum: Groundwater Protection Standards for Municipal Solid Waste Disposal Facilities. Cheyenne, Wyoming. March 15.

Wyoming Department of Environmental Quality (WDEQ), Water Quality Division. 2015. Water Quality Rules & Regulations, Chapter 8, Quality Standards for Wyoming Groundwaters. Cheyenne, WY. September 15.

Revised August 23, 2017

TABLES

**TABLE 1. MONITORING WELL HISTORY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Monitoring Well/Piezometer	Date Drilled	Current Aquifer Designation	Report Reference	Current Status
C-1	1999	Semi-Confined	Hydrogeologic Study 2000, Terracon	Active
C-2	1999	Semi-Confined	Hydrogeologic Study 2000, Terracon	Active
C-3	1999	Semi-Confined	Hydrogeologic Study 2000, Terracon	Active
D-1	1982	Semi-Confined	Environmental Monitoring Plan 2000, IME	P&A, 1999
D-1A	1982	Principal Unconfined	Environmental Monitoring Plan 2000, IME	P&A, 1999
D-2	1982	Semi-Confined	Environmental Monitoring Plan 2000, IME	P&A, 1999
D-2A	1982	Principal Unconfined	Environmental Monitoring Plan 2000, IME	P&A, 1999
D-3	1982	Semi-Confined	Environmental Monitoring Plan 2000, IME	P&A, 1999
D-4	1982	Principal Unconfined	Environmental Monitoring Plan 2000, IME	P&A, 1999
Eagle #2	1988	Unknown	1995 Renewal Permit, Exhibit 20B	Unknown if well exists
Eagle #3	1988	Unknown	1995 Renewal Permit, Exhibit 20B	Unknown if well exists
Eagle #4	1988	Unknown	1995 Renewal Permit, Exhibit 20B	Unknown if well exists
EE-01	1996	Perched 2	Environmental Monitoring Plan 2000, IME	P&A, 1999
EE-02	1996	Perched 2	Environmental Monitoring Plan 2000, IME	P&A, 1999
EE-04	1996	Perched 1	1998 Permit Renewal, ref East End Groundwater Study Feb 1997	Active
EE-05	1996	Perched 1	1998 Permit Renewal, ref East End Groundwater Study Feb 1997	P&A, Scheduled 2017
EE-06	1996	Perched 1	1998 Permit Renewal, ref East End Groundwater Study Feb 1997	P&A, Scheduled 2017
EE-07	1996	Perched 1	1998 Permit Renewal, ref East End Groundwater Study Feb 1997	Active
GMW-01	2013	Semi-Confined	Site Characterization Report, HJ Landfill Expansion 2014, Golder	Active
GMW-03	2013	Semi-Confined	Site Characterization Report, HJ Landfill Expansion 2014, Golder	Active
GMW-04	2013	Semi-Confined	Site Characterization Report, HJ Landfill Expansion 2014, Golder	Active
GMW-06	2013	Semi-Confined	Site Characterization Report, HJ Landfill Expansion 2014, Golder	Active
GMW-07	2013	Semi-Confined	Site Characterization Report, HJ Landfill Expansion 2014, Golder	Active
No. 4	1975	Perched 1	1998 Permit, Exhibit 21	Converted to methane vent in 2014
No. 7	1975	Perched 1	Environmental Monitoring Plan 2000, IME	P&A, 1999
No. 8	1977	Perched 1	1998 Permit, Exhibit 21	P&A, 2015
P-1	1985	Perched 1	1998 Permit, Exhibit 21	Active
P-2	1985	Perched 1	Environmental Monitoring Plan 2000, IME	P&A, 1999
P-3	1985	Perched 1	1998 Permit, Exhibit 21	Active
P-4	1985	Perched 1	1998 Permit, Exhibit 21	Active
P-5	1985	Perched 1	1998 Permit, Exhibit 21	Active
P-7	1990	Leachate	1998 Permit, Exhibit 21	P&A, Scheduled 2017
P-8	1990	Perched 1	1998 Permit, Exhibit 21	Active

TABLE 1. MONITORING WELL HISTORY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING

Monitoring Well/Piezometer	Date Drilled	Current Aquifer Designation	Report Reference	Current Status
P-9	1990	Perched 1	1998 Permit, Exhibit 21	P&A, Scheduled 2017
P-10	1990	Perched 1	1998 Permit, Exhibit 21	Active
P-11	1990	Perched 1	1998 Permit, Exhibit 21	Active
P-12	1990	Perched 1	Environmental Monitoring Plan 2000, IME	P&A, 1999
P1-2	NA	Perched 2	Environmental Monitoring Plan 2000, IME	P&A, 1999
P1E-1	1999	Leachate	Hydrogeologic Study 2000, Terracon	Active
P1E-2	1999	Perched 1	Hydrogeologic Study 2000, Terracon	Abandoned, 2000?
P1E-9	1999	Perched 1	Hydrogeologic Study 2000, Terracon	Active
P3-1	1999	Principal Unconfined	Hydrogeologic Study 2000, Terracon	Active
P3-2	1999	Principal Unconfined	Hydrogeologic Study 2000, Terracon	Active
P3-3	1999	Principal Unconfined	Hydrogeologic Study 2000, Terracon	Active
PU-1	1999	Principal Unconfined	Hydrogeologic Study 2000, Terracon	Active
PU-2	1999	Principal Unconfined	Hydrogeologic Study 2000, Terracon	Active
PU-3	1999	Principal Unconfined	Hydrogeologic Study 2000, Terracon	Active
PU-4	2008	Principal Unconfined	PUA Hydrogeologic Investigation 2009, Trihydro	Active
PU-5	2008	Principal Unconfined	PUA Hydrogeologic Investigation 2009, Trihydro	Active
PU-6	2008	Principal Unconfined	PUA Hydrogeologic Investigation 2009, Trihydro	P&A, Scheduled 2017
S-1	1982	Leachate	1998 Permit, Exhibit 21	Active
S-2	1982	Leachate	1998 Permit, Exhibit 21	P&A, Scheduled 2017
S-3	1982	Leachate	1998 Permit, Exhibit 21	P&A, Scheduled 2017
S-4	1982	Leachate	1998 Permit, Exhibit 21	Active
S-5	1982	Leachate	1998 Permit, Exhibit 21	P&A, Scheduled 2017
S-6	1982	Perched 1	1998 Permit, Exhibit 21	P&A, 12/2009
S-7	1982	Leachate	1998 Permit, Exhibit 21	P&A, Scheduled 2017
S-7A	1982	Leachate	1998 Permit, Exhibit 21	Active
S-8	1982	Leachate	1998 Permit, Exhibit 21	P&A, Scheduled 2017
S-9	1982	Perched 1	1998 Permit, Exhibit 21	Active
S-9A	1982	Perched 1	1998 Permit, Exhibit 21	P&A, Scheduled 2017
S-10	1982	Perched 1	Environmental Monitoring Plan 2000, IME	P&A, 1999
S-11	1982	Perched 1	1998 Permit, Exhibit 21	P&A, 2015
S-11A	1982	Perched 1	1998 Permit, Exhibit 21	Converted to methane vent in 2014
S-11B	1982	Leachate	1998 Permit, Exhibit 21	P&A, Scheduled 2017
S-12	1982	Perched 1	Environmental Monitoring Plan 2000, IME	P&A, 1999
S-13	1982	Perched 1	1998 Permit, Exhibit 21	P&A, Scheduled 2017

**TABLE 1. MONITORING WELL HISTORY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Monitoring Well/Piezometer	Date Drilled	Current Aquifer Designation	Report Reference	Current Status
S-14	1982	Perched 1	Environmental Monitoring Plan 2000, IME	P&A, 1999
SRM-2	1989	Perched 1	1998 Permit, Exhibit 21	P&A, 2015
SRM-4	1990	Principal Unconfined	1998 Permit, Exhibit 21	Active
SRM-5	1990	Principal Unconfined	1998 Permit, Exhibit 21	Active
SRM-6	1990	Perched 1	1998 Permit, Exhibit 21	Active
SRM-7	1990	Principal Unconfined	1998 Permit, Exhibit 21	Active
SRM-8	1990	Principal Unconfined	1998 Permit, Exhibit 21	Active

TABLE 2. CURRENT MONITORING WELL, SUMP, AND PIEZOMETER CONSTRUCTION SUMMARY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING

Revised August 23, 2017

Monitoring Well/ Piezometer	Classification	Hydraulic Location	Casing Inside Diameter	Ground Surface Elevation (ft-amsl)	Top of Casing Elevation (ft-amsl)	Total Depth Aug-16 (ft-btoc)	Depth to Water Aug-16 (ft-btoc)	Last Contained Measurable Water
Semi-Confined Aquifer Wells								
C-1	Detection	Up-gradient	2-inch	6621.289	6623.329	260.00	216.10	Aug-16
C-2	Detection	Down-gradient	2-inch	6548.208	6550.468	259.50	149.60	Aug-16
C-3	Detection	Down-gradient	2-inch	6600.104	6601.416	247.00	198.67	Aug-16
GMW-01	Detection	Down-gradient	2-inch	6596.867	6599.371	218.38	Dry	Drilling, 2013
GMW-03	Detection	Down-gradient	2-inch	6579.348	6581.368	217.17	201.95	Aug-16
GMW-04	Detection	Down-gradient	2-inch	6582.018	6584.19	218.55	204.67	Aug-16
GMW-06	Detection	Down-gradient	2-inch	6560.635	6563.047	188.95	161.51	Aug-16
GMW-07	Detection	Down-gradient	2-inch	6565.371	6567.913	238.53	200.66	Aug-16
Principal Unconfined Aquifer Wells								
PU-1	Detection	Up-gradient	2-inch	6618.595	6620.804	194.59	151.95	Aug-16
PU-2	Detection	Down-gradient	2-inch	6636.789	6638.386	178.89	164.95	Aug-16
PU-3	Detection	Up-gradient	2-inch	6647.535	6649.63	182.95	179.92	Aug-16
PU-4	Detection	Down-gradient	2-inch	6621.354	6623.728	189.75	185.15	Aug-16
PU-5	Detection	Down-gradient	2-inch	6547.811	6550.108	147.95	Dry	Dec-14
P3-1	Detection	Up-gradient	2-inch	6575.244	6577.528	153.25	148.45	Aug-16
P3-2	Detection	Down-gradient	2-inch	6539.942	6542.242	168.32	Dry	Dec-11
P3-3	Detection	Up-gradient	2-inch	6592.15	6595.243	182.84	175.97	Aug-16
SRM-4	Assessment	Down-gradient	5-inch	6558.571	6560.907	174.95	164.95	Aug-16
SRM-5	Assessment	Down-gradient	5-inch	6572.817	6574.097	177.25	158.58	Aug-16
SRM-7	Assessment	Down-gradient	5-inch	6600.087	6601.778	207.45	206.90	Aug-16
SRM-8	Assessment	Down-gradient	5-inch	6619.662	6621.465	203.40	186.18	Aug-16
Perched Water Bearing Zone Wells								
EE-04	Assessment	Down-gradient	2-inch	6558.006	6560.34	58.68	48.71	Aug-16
EE-07	Assessment	Down-gradient	2-inch	6547.747	6550.295	76.41	Dry	Aug-15
P-1	Detection	Up-gradient	2-inch	6603.55	6605.263	12.00	Dry	Apr-16
P-3	Detection	Up-gradient	2-inch	6599.325	6598.073	15.10	11.90	Aug-16
P-4	Detection	Up-gradient	2-inch	6592.134	6592.02	13.55	12.80	Aug-16
P-5	Detection	Up-gradient	2-inch	6594.964	6597.02	10.87	8.55	Aug-16
P-8	Detection	Up-gradient	2-inch	6591.297	6593.121	9.10	Dry	Apr-16
P-10	Detection	Up-gradient	2-inch	6589.189	6591.331	10.65	Dry	Apr-16
P-11	Detection	Up-gradient	2-inch	6586.945	6589.826	7.52	Dry	Apr-16
P1E-9	Assessment	Down-gradient	2-inch	6557.593	6559.985	55.45	50.70	Aug-16
S-9	Detection	Up-gradient	2-inch	6595.177	6596.493	15.57	Dry	Aug-15
SRM-6	Assessment	Down-gradient	5-inch	6574.928	6577.524	65.05	46.53	Aug-16

TABLE 2. CURRENT MONITORING WELL, SUMP, AND PIEZOMETER CONSTRUCTION SUMMARY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING

Revised August 23, 2017

Monitoring Well/ Piezometer	Classification	Hydraulic Location	Casing Inside Diameter	Ground		Top of Casing Elevation (ft-amsl)	Total Depth Aug-16 (ft-btoc)	Depth to Water Aug-16 (ft-btoc)	Last Contained Measurable Water
				Surface Elevation (ft-amsl)					
Leachate Wells									
P1E-1	Leachate	NA	2-inch	6589.566	6591.955	62.37	60.48		Aug-16
S-1	Leachate	NA	3-inch	6615.051	6617.73	23.55	Dry		Dec-15
S-4	Leachate	NA	3-inch	6612.125	6613.684	24.61	24.55		Aug-16
S-7A	Leachate	NA	2-inch	6604.273	6607.771	49.53	49.41		Aug-16
Leachate Sumps									
NT-1	Leachate	NA	6-inch	6643.908	6649.061	64.46	Dry		Aug-14
NT-2	Leachate	NA	6-inch	6636.99	6634.98	62.98	60.15		Aug-16
NT-3	Leachate	NA	10-inch	6623.384	6627.984	NM	NM		Aug-16
NT-4	Leachate	NA	10-inch	6601.426	6606.761	NM	NM		Aug-16
NT-5	Leachate	NA	16-inch	6552.013	6552.004	NM	NM		Aug-16

Notes:

ft - feet

ft-amsl - feet above mean sea level

ft-bloc - feet below top of casing

Information in this table was based on the best known data available

NM - not gauged manually (electronic display in use)

**TABLE 3. SAMPLING SCHEDULE SUMMARY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Monitoring Location	Sampling Requirements ¹	
	April	August
Semi-Confined Aquifer Wells		
C-1	Appendix A, C	Appendix C
C-2	Appendix A, C	Appendix C
C-3	Appendix A, C	Appendix C
GMW-01	Appendix A, C	Appendix C
GMW-03	Appendix A, C	Appendix C
GMW-04	Appendix A, C	Appendix C
GMW-06	Appendix A, C	Appendix C
GMW-07	Appendix A, C	Appendix C
Principal Unconfined Aquifer Wells		
PU-1	Appendix A, C	Appendix C
PU-2	Appendix A, C	Appendix C
PU-3	Appendix A, C	Appendix C
PU-4	Appendix A, C	Appendix C
PU-5	Appendix A, C	Appendix C
P3-1	Appendix A, C	Appendix C
P3-2	Appendix A, C	Appendix C
P3-3	Appendix A, C	Appendix C
SRM-4	Appendix B (VOC),C ²	Appendix C
SRM-5	Appendix B (VOC),C ²	Appendix C
SRM-7	Appendix B (VOC),C ²	Appendix C
SRM-8	Appendix B (VOC),C ²	Appendix C
Perched Water Bearing Zone Wells		
EE-04	Appendix B (VOC),C ²	Appendix C
EE-07	Appendix B (VOC),C ²	Appendix C
P1E-9	Appendix B (VOC),C ²	Appendix C
P-5	Appendix A, C	Appendix C
S-9	Appendix A, C	Appendix C
SRM-6	Appendix B (VOC),C ²	Appendix C
P-1	Appendix A, C	Appendix C
P-3	Appendix A, C	Appendix C
P-4	Appendix A, C	Appendix C
P-5	Appendix A, C	Appendix C
P-8	Appendix A, C	Appendix C
P-10	Appendix A, C	Appendix C
P-11	Appendix A, C	Appendix C
Leachate Sumps		
NT-5	Appendix A, C	N/A

Abbreviations:

N/A = Not applicable

Notes:

1. If an annual sample is required and cannot be collected in April, attempt will be made in August.
2. Full Appendix B parameters will be collected once every five years.

**TABLE 4. FLUID MONITORING SCHEDULE SUMMARY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Monitoring Location	Fluid Monitoring Requirements and Event	
	April	August
Semi-Confined Aquifer Wells		
C-1	Required	Required
C-2	Required	Required
C-3	Required	Required
GMW-01	Required	Required
GMW-03	Required	Required
GMW-04	Required	Required
GMW-06	Required	Required
GMW-07	Required	Required
Principal Unconfined Aquifer Wells		
PU-1	Required	Required
PU-2	Required	Required
PU-3	Required	Required
PU-4	Required	Required
PU-5	Required	Required
P3-1	Required	Required
P3-2	Required	Required
P3-3	Required	Required
SRM-4	Required	Required
SRM-5	Required	Required
SRM-7	Required	Required
SRM-8	Required	Required
Perched Water Bearing Zone Wells		
EE-04	Required	Required
EE-07	Required	Required
P1E-9	Required	Required
S-9	Required	Required
SRM-6	Required	Required
P-1	Required	Required
P-3	Required	Required
P-4	Required	Required
P-5	Required	Required
P-8	Required	Required
P-10	Required	Required
P-11	Required	Required

**TABLE 4. FLUID MONITORING SCHEDULE SUMMARY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Monitoring Location	Fluid Monitoring Requirements and Event	
	April	August
Leachate Wells		
P1E-1	Required	Required
S-1	Required	Required
S-4	Required	Required
S-7A	Required	Required
Leachate Sumps¹		
NT-1	Required	Required
NT-2	Required	Required

Note:

1. Fluid levels for NT-3, NT-4 and NT-5 read from electronic display monthly by City staff.

**TABLE 5. WDEQ/SHWD APPENDIX A CONSITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Common name ²	CAS RN ³	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Inorganics (15)				
Antimony	(Total)	Antimony	6010 7040 7041	300 2000 30
Arsenic	(Total)	Arsenic	6010 7060 7061	500 10 20
Barium	(Total)	Barium	6010 7080	20 1000
Beryllium	(Total)	Beryllium	6010 7090 7091	3 50 2
Cadmium	(Total)	Cadmium	6010 7130 7131	40 50 1
Chromium	(Total)	Chromium	6010 7190 7191	70 500 10
Cobalt	(Total)	Cobalt	6010 7200 7201	70 500 10
Copper	(Total)	Copper	6010 7210 7211	60 200 10
Lead	(Total)	Lead	6010 7420 7421	400 1000 10
Nickel	(Total)	Nickel	6010 7520	150 400
Selenium	(Total)	Selenium	6010 7740 7741	750 20 20
Silver	(Total)	Silver	6010 7760	70 100
Thallium	(Total)	Thallium	6010 7840 7841	400 1000 10
Vanadium	(Total)	Vanadium	6010 7910 7911	80 2000 40
Zinc	(Total)	Zinc	6010 7950 7951	20 50 0.5
Volatiles (47)				
Acetone	67-64-1	2-Propanone	8260	100
Acrylonitrile	107-13-1	2-Propenenitrile	8030 8260	5 200
Benzene	71-43-2	Benzene	8020 8021 8260	2 0.1 5
Bromochloromethane; Chlorobromomethane	74-97-5	Methane, bromochloro-	8021 8260	0.1 5
Bromodichloromethane; Dibromochloromethane	75-27-4	Methane, bromodichloro-	8010 8021 8260	1 0.2 5
Bromoform; Tribromomethane	75-25-2	Methane, tribromo-	8010 8021 8260	2 15 5

**TABLE 5. WDEQ/SHWD APPENDIX A CONSITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Common name ²	CAS RN ³	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Carbon disulfide	75-15-0	Carbon disulfide	8260	100
Carbon tetrachloride	56-23-5	Methane, tetrachloro-	8010 8021 8260	1 0.1 10
Chlorobenzene	108-90-7	Benzene, chloro-	8010 8020 8021 8260	2 2 0.1 5
Chloroethane; Ethyl chloride	75-00-3	Ethane, chloro-	8010 8021 8060	5 1 10
Chloroform; Trichloromethane	67-66-3	Methane, trichloro-	8010 8021 8260	0.5 0.2 5
Dibromochloromethane; Chlorodibromomethane	124-48-1	Methane, dibromochloro-	8010 8021 8260	1 0.3 5
1,2-Dibromo-3-chloropropane; DBCP	96-12-8	Propane, 1,2-dibromo-3-chloro-	8011 8021 8260	0.1 30 25
1,2-Dibromoethane; Ethylene dibromide; EDB	106-93-4	Ethane, 1,2-dibromo-	8011 8021	0.1 10
o-Dichlorobenzene	95-50-1	Benzene, 1,2-dichloro-	8010 8020 8021 8120 8260 8270	2 5 0.5 10 5 10
p-Dichlorobenzene; 1,4 Dichlorobenzene	106-46-7	Benzene, 1,4-dichloro-	8010	2
trans-1,4-Dichloro-2-butene	110-57-6	2-Butene, 1,4-dichloro-, (E)-	8260	100
1,1-Dichloroethane; Ethylidene chloride	75-34-3	Ethane, 1,1-dichloro-	8010 8021 8260	1 0.5 5
1,2-Dichloroethane; Ethylene dichloride	107-06-2	Ethane, 1,1-dichloro-	8010 8021 8260	0.5 0.3 5
1,1-Dichloroethylene; 1,1- Dichloroethene; Vinylidene chloride	75-35-4	Ethene, 1,1-dichloro-	8010 8021 8260	1 0.5 5
cis-1,2-Dichloroethylene; cis-1,2- Dichloroethene	156-59-2	Ethene, 1,2-dichloro-, (Z)-	8021 8260	0.2 5
trans-1,2-Dichloroethylene trans-1,2- Dichloroethene	156-60-5	Ethene, 1,2-dichloro-, (E)-	8010 8021 8260	1 0.5 5
1,2-Dichloropropane; Propylene dichloride	78-87-5	Propane, 1,2-dichloro-	8010 8021 8260	0.5 0.05 5
cis-1,3-Dichloropropene	10061-01-5	1-Propene, 1,3-dichloro-, (Z)-	8010 8260	20 10
trans-1,3-Dichloropropene	10061-02-6	1-Propene, 1,3-dichloro-, (E)-	8010	5
Ethylbenzene	100-41-4	Benzene, ethyl-	8020 8221 8260	2 0.05 5
2-Hexanone; Methyl butyl ketone	591-78-6	2-Hexanone	8260	50
Methyl bromide; Bromomethane	74-83-9	Methane, bromo-	8010 8021	20 10
Methyl chloride; Chloromethane	74-87-3	Methane, chloro-	8010 8021	1 0.3

**TABLE 5. WDEQ/SHWD APPENDIX A CONSITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Common name ²	CAS RN ³	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Methylene bromide; Dibromomethane	74-95-3	Methane, dibromo-	8010 8021 8260	15 20 10
Methylene chloride; Dichloromethane	75-09-2	Methane, dichloro-	8010 8021 8260	5 0.2 10
Methyl ethyl ketone; MEK; 2-Butanone	78-93-3	2-Butanone	8015 8260	10 100
Methyl iodide; Iodomethane	74-88-4	Methane, iodo-	8010 8260	40 10
4-Methyl-2-pentanone; Methyl isobutyl ketone	108-10-1	2-Pentanone, 4-methyl-	8015 8260	5 100
Styrene	100-42-5	Benzene, ethenyl-	8020 8021 8260	1 0.1 10
1,1,1,2-Tetrachloroethane	630-20-6	Ethane, 1,1,1,2-tetrachloro-	8010 8021 8260	5 0.05 5
1,1,2,2-Tetrachloroethane	79-34-5	Ethane, 1,1,2,2-tetrachloro-	8010 8021 8260	0.5 0.1 5
Tetrachloroethylene; Tetrachloroethene; Perchloroethylene	127-18-4	Ethene, tetrachloro-	8010 8021 8260	0.5 0.5 5
Toluene	108-88-3	Benzene, methyl-	8020 8021 8260	2 0.1 5
1,1,1-Trichloroethane; Methylchloroform	71-55-6	Ethane, 1,1,1-trichloro-	8010 8021 8260	0.3 0.3 5
1,1,2-Trichloroethane	79-00-5	Ethane, 1,1,2-trichloro-	8010 8260	0.2 5
Trichloroethylene; Trichloroethene	79-01-6	Ethene, trichloro-	8010 8021 8260	1 0.2 5
Trichlorofluoromethane; CFC- 11	75-69-4	Methane, trichlorofluoro-	8010 8021 8260	10 0.3 5
1,2,3-Trichloropropane	96-18-4	Propane, 1,2,3-trichloro-	8010 8021 8260	10 5 15
Vinyl acetate	108-05-4	Acetic acid, ethenyl ester	8260	50
Vinyl chloride; Chloroethene	75-01-4	Ethene, chloro-	8010 8021 8260	2 0.4 10
Xylene (total)	See Note 11	Benzene, dimethyl-	8020 8021 8260	5 0.2 5

Notes:

¹The regulatory requirements pertain only to the list of substances; the right hand columns (Methods and PQL) are given for informational purposes only. See also footnotes 5 and 6.

²Common names are those widely used in government regulations, scientific publications, and commerce; synonyms exist for many chemicals.

³Chemical Abstracts Service registry number. Where "Total" is entered, all species in the groundwater that contain this element are included.

⁴CAS index names are those used in the 9th Collective Index.

⁵Suggested Methods refer to analytical procedure numbers used in EPA Report SW-846 "Test Methods for Evaluating Solid Waste", third edition, November 1986, as revised, December 1987. Analytical details can be found in SW-846 and in documentation on file at the Department. CAUTION: The methods listed are representative SW-846 procedures and may not always be the most suitable method(s) for monitoring an analyte under the regulations.

⁶Practical Quantitation Limits (PQLs) are the lowest concentrations of analytes in groundwaters that can be reliably determined within specified limits of precision and accuracy by the indicated methods under routine laboratory operating conditions. The PQLs listed are generally stated to one significant figure. PQLs are based on 5 mL samples for volatile organics and 1 L samples for semivolatile organics. CAUTION: The PQL values in many cases are based only on a general estimate for the method and not on a determination for individual compounds; PQLs are not a part of the regulation.

**TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Common name ²	CAS RN ³ 10061-02-6	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Inorganics (19)				
Antimony	(Total)	Antimony	6010 7040 7041	300 2000 30
Arsenic	(Total)	Arsenic	6010 7060 7061	500 10 20
Barium	(Total)	Barium	6010 7080	20 1000
Beryllium	(Total)	Beryllium	6010 7090 7091	3 50 2
Cadmium	(Total)	Cadmium	6010 7130 7131	40 50 1
Chromium	(Total)	Chromium	6010 7190 7191	70 500 10
Cobalt	(Total)	Cobalt	6010 7200 7201	70 500 10
Copper	(Total)	Copper	6010 7210 7211	60 200 10
Cyanide	57-12-5	Cyanide	9010	200
Lead	(Total)	Lead	6010 7420 7421	400 1000 10
Mercury	(Total)	Mercury	7470	2
Nickel	(Total)	Nickel	6010 7520	150 400
Selenium	(Total)	Selenium	6010 7740 7741	750 20 20
Silver	(Total)	Silver	6010 7760	70 100
Sulfide	18496-25-8	Sulfide	9030	4000
Thallium	(Total)	Thallium	6010 7840 7841	400 1000 10
Tin	(Total)	Tin	6010	40
Vanadium	(Total)	Vanadium	6010 7910 7911	80 2000 40
Zinc	(Total)	Zinc	6010 7950 7951	20 50 0.5

TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING

Revised August 23, 2017

Common name ²	CAS RN ³ 10061-02-6	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Volatiles (64)				
Acetone	67-64-1	2-Propanone	8260	100
Acetonitrile; Methyl cyanide	75-05-8	Acetonitrile	8015	100
Acrolein	107-02-8	2-Propenal	8030	5
Acrylonitrile	107-13-1	2-Propenenitrile	8260 8030 8260	100 5 200
Allyl chloride	107-05-1	1-Propene, 3-chloro-	8010 8260	5 10
Benzene	71-43-2	Benzene	8020 8021 8260	2 0.1 5
Bromochloromethane; Chlorobromomethane	74-97-5	Methane, bromochloro-	8021 8260	0.1 5
Bromodichloromethane; Dibromochloromethane	75-27-4	Methane, bromodichloro-	8010 8021 8260	1 0.2 5
Bromoform; Tribromomethane	75-25-2	Methane, tribromo-	8010 8021 8260	2 15 5
Carbon disulfide	75-15-0	Carbon disulfide	8260	100
Carbon tetrachloride	56-23-5	Methane, tetrachloro-	8010 8021 8260	1 0.1 10
Chlorobenzene	108-90-7	Benzene, chloro-	8010 8020 8021 8260	2 2 0.1 5
Chloroethane; Ethyl chloride	75-00-3	Ethane, chloro-	8010 8021 8060	5 1 10
Chloroform; Trichloromethane	67-66-3	Methane, trichloro-	8010 8021 8260	0.5 0.2 5
Chloroprene	126-99-8	1,3-Butadiene, 2-chloro-	8010 8260	50 20
Dibromochloromethane; Chlorodibromomethane	124-48-1	Methane, dibromochloro-	8010 8021 8260	1 0.3 5
1,2-Dibromo-3-chloropropane; DBCP	96-12-8	Propane, 1,2-dibromo-3-chloro-	8011 8021 8260	0.1 30 25
1,2-Dibromoethane; Ethylene dibromide; EDB	106-93-4	Ethane, 1,2-dibromo-	8011 8021	0.1 10
o-Dichlorobenzene	95-50-1	Benzene, 1,2-dichloro-	8010 8020 8021 8120 8260 8270	2 5 0.5 10 5 10
m-Dichlorobenzene; 1,3-Dichlorobenzene	541-73-1	Benzene, 1,3-dichloro-	8010 8020 8021 8120 8260 8270	5 5 0.2 10 5 10
p-Dichlorobenzene; 1,4-Dichlorobenzene	106-46-7	Benzene, 1,4-dichloro-	8020 8021 8120 8260 8270	5 0.2 10 5 10
p-Dichlorobenzene; 1,4 Dichlorobenzene	106-46-7	Benzene, 1,4-dichloro-	8010	2
trans-1,4-Dichloro-2-butene	110-57-6	2-Butene, 1,4-dichloro-, (E)-	8260	100
Dichlorodifluoromethane	75-71-8		8260	--
1,1-Dichloroethane; Ethylidene chloride	75-34-3	Ethane, 1,1-dichloro-	8010 8021 8260	1 0.5 5

**TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Common name ²	CAS RN ³ 10061-02-6	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,8}
1,2-Dichloroethane; Ethylene dichloride	107-06-2	Ethane, 1,1-dichloro-	8010 8021 8260	0.5 0.3 5
1,1-Dichloroethylene; 1,1- Dichloroethene; Vinylidene chloride	75-35-4	Ethene, 1,1-dichloro-	8010 8021 8260	1 0.5 5
cis-1,2-Dichloroethylene; cis-1,2-Dichloroethene	156-59-2	Ethene, 1,2-dichloro-, (Z)-	8021 8260	0.2 5
trans-1,2-Dichloroethylene trans-1,2-Dichloroethene	156-60-5	Ethene, 1,2-dichloro-, (E)-	8010 8021 8260	1 0.5 5
1,2-Dichloropropane; Propylene dichloride	78-87-5	Propane, 1,2-dichloro-	8010 8021 8260	0.5 0.05 5
1,3-Dichloropropane; Trimethylene dichloride	142-28-9	Propane, 1,3-dichloro-	8021 8260	0.3 15
2,2-Dichloropropane; Isopropylidene chloride	594-20-7	Propane, 2,2-dichloro-	8021 8260	0.5 5
1,1-Dichloropropene;	563-58-6	<i>1-Propene, 1,1-dichloro-</i>	8021 8260	0.2 5
cis-1,3-Dichloropropene	10061-01-5	<i>1-Propene, 1,3-dichloro-, (Z)-</i>	8010 8260	20 10
trans-1,3-Dichloropropene	10061-02-6	1-Propene, 1,3-dichloro-, (E)-	8010 8260	5 5
Ethylbenzene	100-41-4	Benzene, ethyl-	8020 8221 8260	2 0.05 5
Ethyl methacrylate	97-63-2	2-Propenoic acid, 2-methyl-, ethyl ester	8015 8260 8270	5 10 10
2-Hexanone; Methyl butyl ketone	591-78-6	2-Hexanone	8260	50
Isobutyl alcohol	78-83-1	1-Propanol, 2-methyl-	8015 8240	50 100
Methacrylonitrile	126-98-7	2-Propenenitrile, 2-methyl-	8015 8260	5 100
Methyl bromide; Bromomethane	74-83-9	Methane, bromo-	8010 8021	20 10
Methyl chloride; Chloromethane	74-87-3	Methane, chloro-	8010 8021	1 0.3
Methylene bromide; Dibromomethane	74-95-3	Methane, dibromo-	8010 8021 8260	15 20 10
Methylene chloride; Dichloromethane	75-09-2	Methane, dichloro-	8010 8021 8260	5 0.2 10
Methyl ethyl ketone; MEK; 2-Butanone	78-93-3	2-Butanone	8015 8260	10 100
Methyl iodide; Iodomethane	74-88-4	Methane, iodo-	8010 8260	40 10
Methyl methacrylate	80-62-6	2-Propenoic acid, 2-methyl-, methyl ester	8015 8260	2 30
4-Methyl-2-pentanone; Methyl isobutyl ketone	108-10-1	2-Pentanone, 4-methyl-	8015 8260	5 100
Naphthalene	91-20-3	Naphthalene	8021 8100 8260 8270	0.5 200 5 10
Propionitrile; Ethyl cyanide	107-12-0	Propanenitrile	8015 8260	60 150
Styrene	100-42-5	Benzene, ethenyl-	8020 8021 8260	1 0.1 10
1,1,1,2-Tetrachloroethane	630-20-6	Ethane, 1,1,1,2-tetrachloro-	8010 8021 8260	5 0.05 5

**TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Common name ²	CAS RN ³ 10061-02-6	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
1,1,2,2-Tetrachloroethane	79-34-5	Ethane, 1,1,2,2-tetrachloro-	8010 8021 8260	0.5 0.1 5
Tetrachloroethylene; Tetrachloroethene; Perchloroethylene	127-18-4	Ethene, tetrachloro-	8010 8021 8260	0.5 0.5 5
Toluene	108-88-3	Benzene, methyl-	8020 8021 8260	2 0.1 5
1,2,4-Trichlorobenzene	120-82-1	Benzene, 1,2,4-trichloro-	8021 8120 8260 8270	0.3 0.5 10 10
1,1,1-Trichloroethane; Methylchloroform	71-55-6	Ethane, 1,1,1-trichloro-	8010 8021 8260	0.3 0.3 5
1,1,2-Trichloroethane	79-00-5	Ethane, 1,1,2-trichloro-	8010 8260	0.2 5
Trichloroethylene; Trichloroethene	79-01-6	Ethene, trichloro-	8010 8021 8260	1 0.2 5
Trichlorofluoromethane; CFC- 11	75-69-4	Methane, trichlorofluoro-	8010 8021 8260	10 0.3 5
1,2,3-Trichloropropane	96-18-4	Propane, 1,2,3-trichloro-	8010 8021 8260	10 5 15
Vinyl acetate	108-05-4	Acetic acid, ethenyl ester	8260	50
Vinyl chloride; Chloroethene	75-01-4	Ethene, chloro-	8010 8021 8260	2 0.4 10
Xylene (total)	See Note 11	Benzene, dimethyl-	8020 8021 8260	5 0.2 5

TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING

Revised August 23, 2017

Common name ²	CAS RN ³ 10061-02-6	Chemical abstracts service Index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Semi-Volatiles (108)				
Acenaphthene	83-32-9	Acenaphthylene, 1,2-dihydro-	8100 8270	200 10
Acenaphthylene	208-96-8	Acenaphthylene	8100 8270	200 10
Acetophenone	98-86-2	Ethanone, 1-phenyl-	8270	10
2-Acetylaminofluorene; 2-AAF	53-96-3	Acetamide, N-9H-fluoren-2-yl-	8270	20
4-Aminobiphenyl	92-67-1	[1,1'-Biphenyl]-4-amine	8270	20
Anthracene	120-12-7	Anthracene	8100 8270	200 10
Benzo[a]anthracene; Benzanthracene	56-55-3	Benz[a]anthracene	8100 8270	200 10
Benzo[b]fluoranthene	205-99-2	Benz[e]acephenanthrylene	8100 8270	200 10
Benzo[k]fluoranthene	207-08-9	Benzo[k]fluoranthene	8100 8270	200 10
Benzo[ghi]perylene	191-24-2	Benzo[ghi]perylene	8100 8270	200 10
Benzo[a]pyrene	50-32-8	Benzo[a]pyrene	8100 8270	200 10
Benzyl alcohol	100-51-6	Benzenemethanol	8270	20
Bis(2-chloroethoxy)methane	111-91-1	Ethane, 1,1'-[methylenebis (oxy)]bis[2-chloro-	8110 8270	5 10
Bis(2-chloroethyl)ether; Dichloroethyl ether	111-44-4	Ethane, 1,1'-oxybis[2-chloro-	8110 8270	3 10
Bis(2-chloro-1-methylethyl) ether; 2,2'- Dichlorodisopropyl ether; DCIP, See note 7	108-60-1	Propane, 2,2'-oxybis[1-chloro-	8110 8270	10 10
Bis(2-ethylhexyl) phthalate	117-81-7	1,2-Benzenedicarboxylic acid, bis(2- ethylhexyl)ester	8060	20
4-Bromophenyl phenyl ether	101-55-3	Benzene, 1-bromo-4-phenoxy-	8110 8270	25 10
Butyl benzyl phthalate; Benzyl butyl phthalate	85-68-7	1,2-Benzenedicarboxylic acid, butyl phenylmethyl ester	8060 8270	5 10
p-Chloroaniline	106-47-8	Benzenamine, 4-chloro-	8270	20
Chlorobenzilate	510-15-6	Benzenecarboxylic acid, 4-chloro- α-(4- chlorophenyl)-α-hydroxy-, ethyl ester	8270	10
p-Chloro-m-cresol; 4-Chloro-3-methylphenol	59-50-7	Phenol, 4-chloro-3-methyl-	8040 8270	5 20
2-Chloronaphthalene	91-58-7	Naphthalene, 2-chloro-	8120 8270	10 10
2-Chlorophenol	95-57-8	Phenol, 2-chloro-	8040 8270	5 10
4-Chlorophenyl phenyl ether	7005-72-3	Benzene, 1-chloro-4-phenoxy-	8110 8270	40 10
Chrysene	218-01-9	Chrysene	8100 8270	200 10
m-Cresol; 3-methylphenol	108-39-4	Phenol, 3-methyl-	8270	10
o-Cresol; 2-methylphenol	95-48-7	Phenol, 2-methyl-	8270	10
p-Cresol; 4-methylphenol	106-44-5	Phenol, 4-methyl-	8270	10
Diallate	2303-16-4	Carbamothioic acid, bis(1- methylethyl)-, S- (2,3- dichloro-2- propenyl) ester	8270	10
Dibenz[a,h]anthracene	53-70-3	Dibenz[a,h]anthracene	8100 8270	200 10
Dibenzofuran	132-64-9	Dibenzofuran	8270	10
3,3'-Dichlorobenzidine	91-94-1	[1,1'-Biphenyl]-4,4'-diamine, 3,3'- dichloro-	8270	20
2,4-Dichlorophenol	120-83-2	Phenol, 2,4-dichloro-	8040 8270	5 10
2,6-Dichlorophenol	87-65-0	Phenol, 2,6-dichloro-	8270	10
Diethyl phthalate	84-66-2	1,2-Benzenedicarboxylic acid, diethyl ester	8060 8270	5 10

TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING

Revised August 23, 2017

Common name ²	CAS RN ³ 10061-02-6	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
O,O-Diethyl O-2-pyrazinyl phosphorothioate;Thionazin	297-97-2	Phosphorothioic acid, O,O- diethyl O-pyrazinyl ester	8141 8270	5 20
Dimethoate	60-51-5	Phosphorodithioic acid, O,O- dimethyl S-[2-(methylamino)-2- oxoethyl] ester	8141 8270	3 20
p-(Dimethylamino)azobenzene	60-11-7	Benzenamine, N,N-dimethyl-4- (phenylazo)-	8270	10
7,12- Dimethylbenz[a]anthracene	57-97-6	Benz[a]anthracene, 7,12- dimethyl-	8270	10
3,3'-Dimethylbenzidine	119-93-7	[1,1'-Biphenyl]-4,4'-diamine, 3,3'- dimethyl-	8270	10
2,4-Dimethylphenol; m-Xylenol	105-67-9	Phenol, 2,4-dimethyl-	8040 8270	5 10
Dimethyl phthalate	131-11-3	1,2-Benzenedicarboxylic acid, dimethyl ester	8060 8270	5 10
m-Dinitrobenzene	99-65-0	Benzene, 1,3-dinitro-	8270	20
4,6-Dinitro-o-cresol; 4,6-Dinitro-2-methylphenol	534-52-1	Phenol, 2-methyl-4,6-dinitro-	8040 8270	150 50
2,4-Dinitrophenol	51-28-5	Phenol, 2,4-dinitro-	8040 8270	150 50
2,4-Dinitrotoluene	121-14-2	Benzene, 1-methyl-2,4-dinitro-	8090 8270	0.2 10
Di-n-butyl phthalate	84-74-2	1,2-Benzenedicarboxylic acid, dibutyl ester	8060 8270	5 10
2,6-Dinitrotoluene	606-20-2	Benzene, 2-methyl-1,3-dinitro-	8090 8270	0.1 10
Dinoseb; DNBP; 2-sec-Butyl-4,6-dinitrophenol	88-85-7	Phenol, 2-(1-methylpropyl)- 4,6-dinitro-	8150 8270	1 20
Di-n-octyl phthalate	117-84-0	1,2-Benzenedicarboxylic acid, dioctyl ester	8060 8270	30 10
Diphenylamine	122-39-4	Benzenamine, N-phenyl-	8270	10
Disulfoton	298-04-4	Phosphorodithioic acid, O,O- diethyl S-[2- (ethylthio)ethyl]ester	8140 8141 8270	2 0.5 10
Ethyl methanesulfonate	62-50-0	Methanesulfonic acid, ethyl ester	8270	20
Famphur	52-85-7	Phosphorothioic acid, O-[4- [(dimethylamino)sulfonyl]phenyl]- O,O-dimethyl ester	8270	20
Fluoranthene	206-44-0	Fluoranthene	8100 8270	200 10
Fluorene	86-73-7	9H-Fluorene	8100 8270	200 10
Hexachlorobenzene	118-74-1	Benzene, hexachloro-	8120 8270	0.5 10
Hexachlorobutadiene	87-68-3	1,3-Butadiene, 1,1,2,3,4,4- hexachloro-	8021 8120 8260 8270	0.5 5 10 10
Hexachlorocyclopentadiene	77-47-4	1,3-Cyclopentadiene, 1,2,3,4,5,5-hexachloro-	8120 8270	5 10
Hexachloroethane	67-72-1	Ethane, hexachloro-	8120 8260 8270	0.5 10 10
Hexachloropropene	1888-71-7	1-Propene, 1,1,2,3,3,3- hexachloro-	8270	10
Indeno(1,2,3-cd)pyrene	193-39-5	Indeno[1,2,3-cd]pyrene	8100 8270	200 10
Isodrin	465-73-6	1,4,5,8- Dimethanonaphthalene,1,2,3,4,1 0,10-hexachloro-1,4,4a,5,8,8a hexahydro- (1α,4α,4aβ,5β,8β,8aβ)-	8270 8260	20 10
Isophorone	78-59-1	2-Cyclohexen-1-one, 3,5,5- trimethyl-	8090 8270	60 10

TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING

Revised August 23, 2017

Common name ²	CAS RN ³	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Isosafrole	10061-02-6	1,3-Benzodioxole, 5-(1-propenyl)-	8270	10
Kepone	143-50-0	1,3,4-Metheno-2H-cyclobuta-[cd]pentalen-2-one, 1,1a,3,3a,4,5,5a,5b,6-decachloro-octahydro-	8270	20
Methapyrillene	91-80-5	1,2-Ethanediamine, N,N-dimethyl-N'-2-pyridinyl-N'-(2-thienylmethyl)-	8270	100
3-Methylcholanthrene	56-49-5	Benz[<i>h</i>]aceanthrylene, 1,2-dihydro-3-methyl-	8270	10
Methyl methanesulfonate	66-27-3	Methanesulfonic acid, methyl ester	8270	10
2-Methylnaphthalene	91-57-6	Naphthalene, 2-methyl-	8270	10
Methyl parathion;	298-00-0	Phosphorothioic acid, O,O-dimethyl O-(4-nitrophenyl)	8140	0.5
Parathion methyl		ester	8141	1
			8270	10
1,4-Naphthoquinone	130-15-4	1,4-Naphthalenedione	8270	10
1-Naphthylamine	134-32-7	1-Naphthalenamine	8270	10
2-Naphthylamine	91-59-8	2-Naphthalenamine	8270	10
o-Nitroaniline; 2-Nitroaniline	88-74-4	Benzenamine, 2-nitro-	8270	50
m-Nitroaniline; 3-Nitroaniline	99-09-2	Benzenamine, 3-nitro-	8270	50
p-Nitroaniline; 4-Nitroaniline	100-01-6	Benzenamine, 4-nitro-	8270	50
Nitrobenzene	98-95-3	Benzene, nitro-	8090	40
			8270	10
o-Nitrophenol; 2-Nitrophenol	88-75-5	Phenol, 2-nitro-	8040	5
			8270	10
p-Nitrophenol; 4-Nitrophenol	100-02-7	Phenol, 4-nitro-	8040	10
			8270	50
N-Nitrosodiethylamine	55-18-5	Ethanamine, N-ethyl-N-nitroso-	8270	20
N-Nitrosodimethylamine	62-75-9	Methanamine, N-methyl-N-nitroso-	8070	2
N-Nitrosodi-n-butylamine	924-16-3	1-Butanamine, N-butyl-N-nitroso-	8270	10
N-Nitrosodiphenylamine	86-30-6	Benzenamine, N-nitroso-N-phenyl-	8070	5
N-Nitrosodipropylamine;	621-64-7	1-Propanamine, N-nitroso-N-propyl-	8070	10
N-Nitroso-N-dipropylamine; Di-n-propylnitrosamine				
N-Nitrosomethylethylamine	10595-95-6	Ethanamine, N-methyl-N-nitroso-	8270	10
N-Nitrosomorpholine	59-89-2	Morpholine, 4-nitroso-	8270	10
N-Nitrosopiperidine	100-75-4	Piperidine, 1-nitroso-	8270	20
N-Nitrosopyrrolidine	930-55-2	Pyrrolidine, 1-nitroso-	8270	40
5-Nitro-o-toluidine	99-55-8	Benzenamine, 2-methyl-5-nitro-	8270	10
Pentachlorophenol	87-86-5	Phenol, pentachloro-	8040	5
			8270	50
Phenanthrene	85-01-8	Phenanthrene	8100	200
			8270	10
Phenol	108-95-2	Phenol	8040	1
p-Phenylenediamine	106-50-3	1,4-Benzenediamine	8270	10
Pentachlorobenzene	608-93-5	Benzene, pentachloro-	8270	10
Pentachloronitrobenzene	82-68-8	Benzene, pentachloronitro-	8270	20
Phenacetin	62-44-2	Acetamide, N-(4-ethoxyphenyl)	8270	20

**TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Common name ²	CAS RN ³	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Phorate	298-02-2	Phosphorodithioic acid, O,O-diethyl S-[(ethylthio)methyl] ester	8140 8141 8270	2 0.5 10
Pronamide	23950-58-5	Benzamide, 3,5-dichloro-N- (1,1-dimethyl-2-propynyl)-	8270	10
Pyrene	129-00-0	Pyrene	8100 8270	200 10
Safrole	94-59-7	1,3-Benzodioxole, 5-(2-propenyl)-	8270	10
1,2,4,5-Tetrachloro-benzene	95-94-3	Benzene, 1,2,4,5-tetrachloro-	8270	10
2,3,4,6-Tetrachlorophenol	58-90-2	Phenol, 2,3,4,6-tetrachloro-	8270	10
o-Toluidine	95-53-4	Benzenamine, 2-methyl-	8270	10
2,4,5-Trichlorophenol	95-95-4	Phenol, 2,4,5-trichloro-	8270	10
2,4,6-Trichlorophenol	88-06-2	Phenol, 2,4,6-trichloro-	8040 8270	5 10
O,O,O-Triethyl phosphorothioate	126-68-1	Phosphorothioic acid, O,O,O-triethyl ester	8270	10
sym-Trinitrobenzene	99-35-4	Benzene, 1,3,5-trinitro-	8270	10

TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING

Revised August 23, 2017

Common name ²	CAS RN ³ 10061-02-6	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Pesticides (20)				
Aldrin	309-00-2	1,4:5,8-Dimethanonaphthalene, 1,2,3,4,10,10-hexachloro- 1,4,4a,5,8,8a-hexahydro- (1α,4α,4aβ,5α, 8α,8aβ)-	8080 8270	0.05 10
alpha-BHC	319-84-6	Cyclohexane, 1,2,3,4,5,6- hexachloro-, (1α, 2α,3β,4α,5β,6β)-	8080 8270	0.05 10
beta-BHC	319-85-7	Cyclohexane, 1,2,3,4,5,6- hexachloro- , (1α,2β,3α,4β,5α,6β)-	8080 8270	0.05 20
delta-BHC	319-86-8	Cyclohexane, 1,2,3,4,5,6- hexachloro-, (1α, 2α,3α,4β,5α,6β)-	8080 8270	0.1 20
gamma-BHC; Lindane	58-89-9	Cyclohexane, 1,2,3,4,5,6- hexachloro- , (1α,2α,3β,4α,5α,6β)-	8080 8270	0.05 20
Chlordane	See Note 8	4,7-Methano-1H-indene, 1,2,4,5,6,7,8,8-octachloro- 2,3,3a,4,7,7a-hexahydro-	8080 8270	0.1 50
4,4'-DDD	72-54-8	Benzene 1,1'-(2,2- dichloroethylidene)bis[4- chloro-	8080 8270	0.1 10
4,4'-DDE	72-55-9	Benzene, 1,1'-(dichloroethenylidene)bis[4- chloro-	8080 8270	0.05 10
4,4'-DDT	50-29-3	Benzene, 1,1'-(2,2,2- trichloroethylidene)bis[4- chloro-	8080 8270	0.1 10
Dieldrin	60-57-1	2,7:3,6-Dimethanonaphth[2,3- b]oxirene, 3,4,5,6,9,9- hexa, chloro- 1a,2,2a,3,6,6a,7,7a-octahydro- , (1α,2β,2αα,3β,6β,6αα,7β,7αα)-	8080 8270	0.05 10
Endosulfan I	959-98-8	6,9-Methano-2,4,3- benzodioxathiepin, 6,7,8,9,10,10-hexachloro- 1,5,5a,6,9,9a-hexahydro-, 3- oxide, (3α,5aβ,6α,9α,9aβ)-	8080 8250	0.1 10
Endosulfan II	33213-65-9	6,9-Methano-2,4,3- benzodioxathiepin, 6,7,8,9,10,10-hexachloro- 1,5,5a,6,9,9a-hexahydro-, 3- oxide, (3α,5αα,6β,9β,9αα)-	8080 8270	0.05 20
Endosulfan sulfate	1031-07-8	6,9-Methano-2,4,3- benzodioxathiepin, 6,7,8,9,10,10-hexachloro- 1,5,5a,6,9,9a-hexahydro-, 3,3- dioxide	8080 8270	0.5 10
Endrin	72-20-8	2,7:3,6-Dimethanonaphth[2,3- b]oxirene, 3,4,5,6,9,9- hexachloro- 1a,2,2a,3,6,6a,7,7a-octahydro- , (1α, 2β,2aβ,3α,6α, 6aβ,7β,7αα)-	8080 8270	0.1 20
Endrin aldehyde	7421-93-4	1,2,4- Methenocyclopenta[cd]pentalene -5-carboxaldehyde, 2,2a,3,3,4,7- hexachlorodecahydro-, (1α,2β,2aβ,4β,4aβ,5β,6β,,6bβ,7 R*)-	8080 8270	0.2 10

**TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Common name ²	CAS RN ³	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Heptachlor	76-44-8	4,7-Methano-1H-indene, 1,4,5,6,7,8-heptachloro-3a,4,7,7a-tetrahydro-	8080 8270	0.05 10
Heptachlor epoxide	1024-57-3	2,5-Methano-2H-indeno[1,2-b]oxirene, 2,3,4,5,6,7,7-heptachloro-1a,1b,5a,6,6a,-hexahydro-, (1aα,1bβ,2α,5α,5aβ,6β,6aα)	8080 8270	1 10
Methoxychlor	72-43-5	Benzene, 1,1'-(2,2,2-trichloroethylidene)bis [4-methoxy-	8080 8270	2 10
Parathion	56-38-2	Phosphorothioic acid, O,O-diethyl-O-(4-nitrophenyl) ester	8141 8270	0.5 10
Toxaphene	See Note 10	Toxaphene	8080	2
Herbicides (3)				
2,4-D; 2,4-Dichlorophenoxy-acetic acid	94-75-7	Acetic acid, (2,4- dichlorophenoxy)-	8150	10
2,4,5-T; 2,4,5-Trichlorophenoxyacetic acid	93-76-5	Acetic acid, (2,4,5-trichlorophenoxy)-	8150	2
Silvex; 2,4,5-TP	93-72-1	Propanoic acid, 2-(2,4,5-trichlorophenoxy)-	8150	2
PCBs (7)				
Polychlorinated biphenyls; PCBs; Aroclors	See Note 9	1,1'-Biphenyl, chloro derivatives	8080 8270	50 200

Notes:

¹The regulatory requirements pertain only to the list of substances; the right hand columns (Methods and PQL) are given for informational purposes only. See also footnotes 5 and 6.

²Common names are those widely used in government regulations, scientific publications, and commerce; synonyms exist for many chemicals.

³Chemical Abstracts Service registry number. Where "Total" is entered, all species in the groundwater that contain this element are included.

⁴CAS index names are those used in the 9th Collective Index.

⁵Suggested Methods refer to analytical procedure numbers used in EPA Report SW-846 "Test Methods for Evaluating Solid Waste", third edition, November 1986, as revised, December 1987. Analytical details can be found in SW-846 and in documentation on file at the Department. CAUTION: The methods listed are representative SW-846 procedures and may not always be the most suitable method(s) for monitoring an analyte under the regulations.

⁶Practical Quantitation Limits (PQLs) are the lowest concentrations of analytes in groundwaters that can be reliably determined within specified limits of precision and accuracy by the indicated methods under routine laboratory operating conditions. The PQLs listed are generally stated to one significant figure. PQLs are based on 5 mL samples for volatile organics and 1 L samples for semivolatile organics. CAUTION: The PQL values in many cases are based only on a general estimate for the method and not on a determination for individual compounds; PQLs are not a part of the regulation.

⁷This substance is often called Bis(2-chloroisopropyl) ether, the name Chemical Abstracts Service applies to its noncommercial isomer, Propane, 2,2'-oxybis[2-chloro- (CAS RN 39638-32-9)]

⁸Chlordane: This entry includes alpha-chlordane (CAS RN 5103-71-9), beta-chlordane (CAS RN 5103-74-2), gamma-chlordane (CAS RN 5566-34-7), and constituents of chlordane (CAS RN 57-74-9 and CAS RN 12789-03-6). PQL shown is for technical chlordane. PQLs of specific isomers are about 20 ug/L by method 8270.

⁹Polychlorinated biphenyls (CAS RN 1336-36-3); this category contains congener chemicals, including constituents of Aroclor 1016 (CAS RN 12674-11-2), Aroclor 1221 (CAS RN 11104-28-2), Aroclor 1232 (CAS RN 11141-16-5), Aroclor 1242 (CAS RN 53469-21-9), Aroclor 1248 (CAS RN 12672-29-6), Aroclor 1254 (CAS RN 11097-69-1), and Aroclor 1260 (CAS RN 11096-82-5). The PQL shown is an average value for PCB congeners.

¹⁰Toxaphene: This entry includes congener chemicals contained in technical toxaphene (CAS RN 8001-35-2), i.e., chlorinated camphene.

¹¹Xylene (total): This entry includes o-xylene (CAS RN 96-47-6), m-xylene (CAS RN 108-38-3), p-xylene (CAS RN. 106-42-3), and unspecified xylenes (dimethylbenzenes) (CAS RN 1330-20-7). PQLs for method 8021 are 0.2 for o-xylene, and 0.1 for m- or p-xylene. The PQL for m-xylene is 2.0 ug/L by method 8020 or 8260.

**TABLE 7. WDEQ/SHWD APPENDIX C CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Constituents	Analytical Method
<u>Constituents for Detection Monitoring</u>	
pH ¹	
Ammonia as N ¹	350.1
Chloride ¹	300.0A
Manganese ¹	SW 6010B
Sulfate ¹	300.0A
Total Dissolved Solids (TDS) ¹	SM 2540C
Bicarbonate ²	SM 2320B
Fluoride ¹	300.0A
Nitrate as N ¹	EPA 353.2
Chemical Oxygen Demand (COD) ²	410.4
Calcium ²	SW 6010B
Iron ¹	SW 6010B
Potassium ²	SW 6010B
Total Organic Carbon (TOC) ²	SM 5310B
Carbonate ²	SM 2320B
Magnesium ²	SW 6010B
Sodium ²	SW 6010B

¹ Constituents that may have a class of use based limit in Chapter 8 of the Wyoming Water Quality Rules and Regulations and/or an MCL.

² Constituents that may be used to characterize and compare groundwater quality. These constituents are useful in determining the similarities and/or differences in the composition of water from specific hydrogeologic units and may help show whether particular units are hydraulically separate or connected. These constituents may be used to classify natural waters and help differentiate between natural variability and a release from a landfill.

**TABLE 8. GROUNDWATER WELLS PERMITTED FOR DOMESTIC OR STOCK WATERING PURPOSES
WITHIN 1/2 MILE OF EXPANSION AREA,
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

SEO PERMIT Number	Permittee	Application Year	Permittee Address	Well Depth (ft-bgs)	Permitted Well Usage
22711	Gary J. & Linda Jo Denney	1973	1466 Happy Jack Road	440	Domestic
112671	Margaret M. Grayer	1998	1498 Happy Jack Road	450	Domestic
49990	Susan E. & Ellis E. Hart	1979	1010 Bade Road	315	Domestic; Stock
179101	William S. Walker	1995	1497 Happy Jack Road.	260	Domestic; Stock
36775	Carl Bade	1977	1016 Bade Road	195	Domestic; Stock
290	Board of Public Utilities	1946	Koppes No. 6	270	Municipal Production
23571	City of Cheyenne	1973	Happy Jack Landfill	300	Miscellaneous
43659	Chris C. and Teresa A. Bade	1978	3703 Charles St	215	Domestic; Stock
231	Arthur L. King	1905	PO BOX 905	180	Domestic; Stock
74560	Board of Public Utilities	1986	PO Box 1469	15	Stock

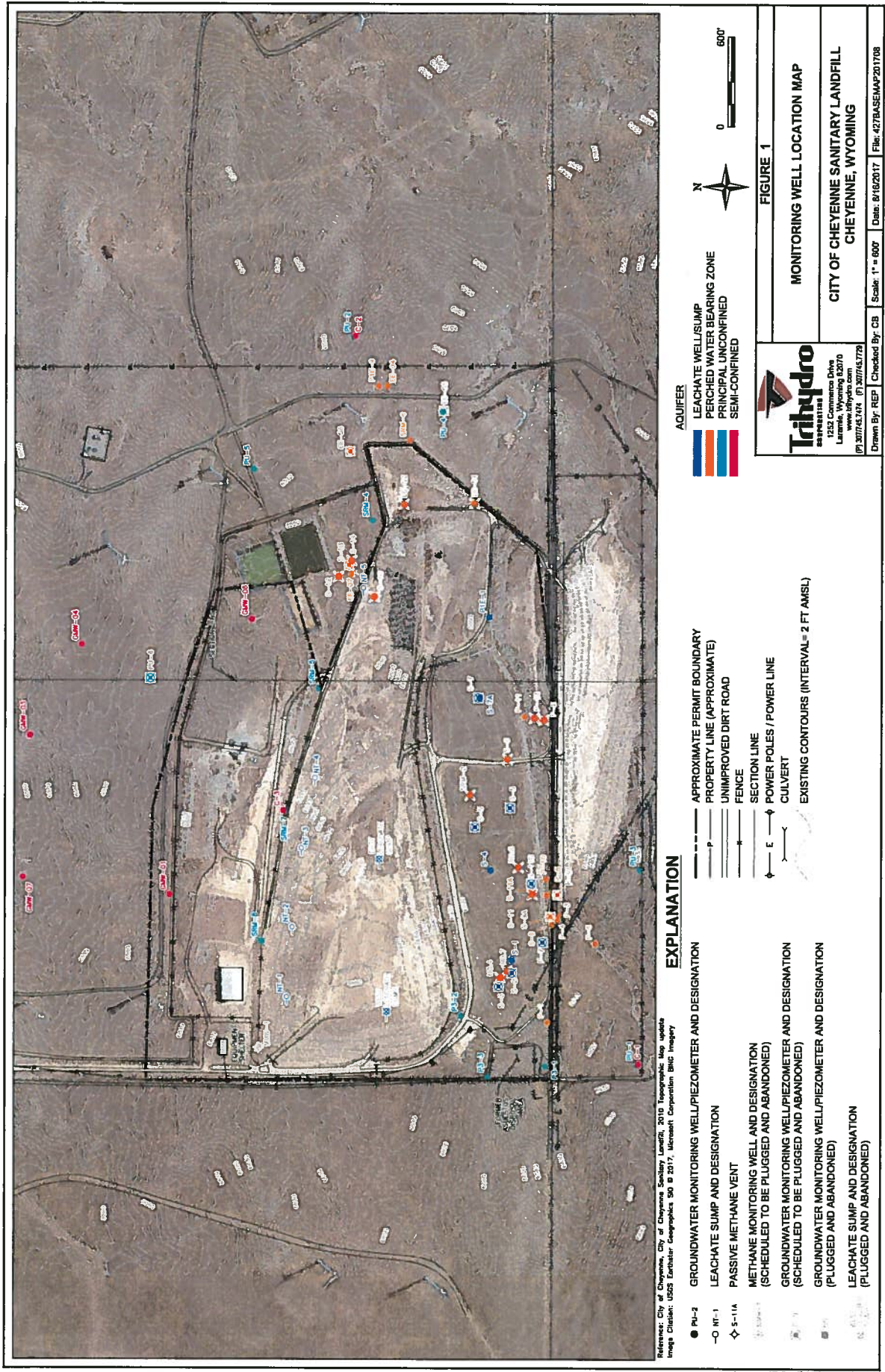
Notes:

Data derived from the Wyoming State Engineer's Office Website, December 12, 2016
ft-bgs - feet below ground surface

**TABLE 9. BACKGROUND AND COMPLIANCE MONITORING WELLS
FOR INTERWELL STATISTICAL COMPARISONS
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Hydrogeologic Zone	Background (Up-gradient) Monitoring Wells	Compliance (Down-gradient) Monitoring Wells
Semi-Confined Aquifer	C-1	C-2
		C-3
		GMW-03
		GMW-04
		GMW-06
Principal Unconfined Aquifer	PU-1 PU-3 P3-1 P3-3	GMW-07
		SRM-4
		SRM-5
		SRM-7
		SRM-8
		PU-2
		PU-4
Perched 1 Water Bearing Zone	P-5 S-9	SRM-6
		EE-04

FIGURE





APPENDIX H-2

ADDENDUM TO THE SANITARY LANDFILL ENVIRONMENTAL MONITORING PLAN (SLEMP) METHANE GAS SAMPLING AND ANALYSIS PLAN

Happy Jack Landfill, Cheyenne, Wyoming

Submitted to: City of Cheyenne
2101 O'Neil Avenue
Cheyenne, Wyoming 82001

Submitted by: Golder Associates Inc.
44 Union Boulevard, Suite 300
Lakewood, Colorado 80228

April 14, 2017

123-81971





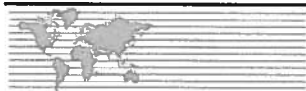


Table of Contents

1.0	INTRODUCTION.....	1
2.0	METHANE GAS MONITORING	2
3.0	REPORTING	4
3.1	Notification of Methane Gas Exceedance.....	4
3.2	Groundwater and Methane Gas Monitoring Report.....	4

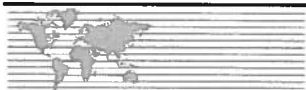
List of Figures

Figure H-2-1 Perimeter Gas Probe Location Map
Figure H-2-2 Typical Gas Probe Detail

List of Appendices

Appendix H-2-1 Field Form – Landfill Gas Monitoring Results





1.0 INTRODUCTION

A Methane Gas Sampling and Analytical Plan (Plan) for the Happy Jack Landfill is required by Chapter 2, Section 9 of the Wyoming Solid Waste Rules and Regulations (Regulations) promulgated by the Wyoming Department of Environmental Quality, Solid and Hazardous Waste Division (WDEQ SHWD). Chapter 2, Section 12(c)(vi) requires the environmental monitoring to continue during closure and post-closure. Hence, this Plan has been developed to guide landfill gas monitoring during both operation and the closure and post-closure periods. Happy Jack Landfill (herein referred to as Landfill) is currently required to conduct methane monitoring, as mandated by Chapter 2, Section 9(c)(i). This Plan provides the proposed locations for methane gas monitoring probes (methane gas network) and the protocols for measuring methane within these probes.

This plan was prepared as an Addendum to the Sanitary Landfill Environmental Monitoring Plan (SLEMP) prepared by Trihydro Corporation in December 2016. The purpose of this Addendum to the SLEMP is to address methane monitoring at the facility boundary. Refer to the SLEMP for information regarding groundwater sampling and analysis, and methane monitoring within on-site structures.



2.0 METHANE GAS MONITORING

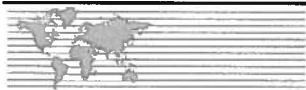
To comply with the requirements of WDEQ SHWD Chapter 2, Section 9(c), methane gas will be monitored at least quarterly in the gas monitoring probes located along the facility boundary, as shown in Figure H-2-1. The methane probes shall be constructed and installed prior to waste filling in Phase 2 in general accordance with the design shown in Figure H-2-2, which provides the critical construction specifications for this permit level submittal. The gas monitoring probes are sited to be spaced at a maximum spacing of 500 feet along the new facility boundary adjacent to the north, east, and west of Phase 2 in accordance with Variance Condition #5. The depth of all probes will coincide with the depth of the landfill in the area adjacent to each probe. This depth will be approximately 50 to 60 feet along the new facility boundary of Phase 2 where the new probes are being installed. Prior to installation, the proposed depth of each probe will be provided to the WDEQ SHWD. It is not anticipated that probes will have to be nested unless there is a technical reason a single screened interval is insufficient to provide monitoring of gas migration at the facility boundary. In the event that a gas probe is damaged, the gas probe will be repaired or replaced as soon as is practical depending on weather conditions and the amount of time required to contract a well driller to install the probe.

Methane gas has historically been monitored in methane monitoring well MMW-1, located south of the equipment building as shown in Figure H-2-1. The city requested to abandon methane monitoring well MMW-1 in 2017.

An As-built Report for the methane probes will be submitted as an addendum to this plan within 60 days of completed construction of each methane probe. This report will include probe construction, boring logs, and a brief narrative and a diagram of each probe. The description of each probe will include probe ID, probe total depth, screened interval, and the corresponding depth of waste in the landfill at the closest point. The report will also include a map showing the location and identification of new, existing, and abandoned gas probes and location coordinates for the probes in State Plane Coordinates (NAD 1983).

Methane gas monitoring is required at least once per quarter at the facility boundary during operation and until the end of the post-closure period. Methane monitoring procedures shall meet the general requirements as outlined below:

- A portable methane meter shall be used for monitoring that is appropriate for the range of methane gas concentrations that are expected. This will generally be methane concentrations that are in the 1% to 5% volume/volume range and can read in both percent volume per volume and percent of the lower explosive limit (LEL).
- The methane meter shall be calibrated to meet the manufacturer's requirements. The calibration data from each monitoring event shall be recorded in the designated project field notebook and/or on a field data sheet.
- The methane meter shall be allowed to warm up and ambient air readings shall be taken at each monitoring location prior to sampling in a probe.



- The methane meter's hose should be connected to the probe using a quick-connect coupling and the meter should be allowed to pump gas from the interior of the probe for approximately five minutes. When a stable reading is achieved, the date, time, and meter readings shall be recorded in the designated project field notebook and/or on a field data sheet in both percent volume by volume and percent of the LEL.

Temperature and initial and final barometric pressure for each monitoring event, along with percent oxygen and percent carbon dioxide, will be recorded on a form comparable to the Field Form shown in Appendix H-2-1. Monitoring records to be maintained on site include the calibration data from each event and all relevant monitoring data. The calibration record will include the meter's manufacturer, model number, and most recent calibration date and data.

Methane monitoring is to continue on a quarterly basis throughout the closure and post-closure period. In order to terminate the post-closure period, a detailed analysis (graphical or statistical) of all available methane monitoring data, including at least 12 consecutive months or 3 consecutive years of quarterly monitoring data at the end of the post-closure period, will be included with the petition to the WDEQ SHWD requesting to terminate the post-closure period. The analysis of methane monitoring data is to demonstrate that methane is not migrating beyond the facility boundary at concentrations greater than 100% of the LEL, and that methane generation rates have stabilized or begun to decline. Abandoned methane monitoring probe boreholes shall be plugged and sealed using a method approved by the WDEQ SHWD.



3.0 REPORTING

3.1 Notification of Methane Gas Exceedance

If the concentration of methane gas at the facility boundary exceeds the LEL, the landfill manager shall:

- Notify the WDEQ SHWD administrator immediately and take steps to protect human health.
- Place a notice in the operating record within seven days of detection, including a copy of the methane test data and a written description of the steps taken to protect human health.
- Implement a remediation plan within 60 days of detection. The plan shall be approved by the WDEQ SHWD and copy kept on file in the operating record.

3.2 Groundwater and Methane Gas Monitoring Report

Methane monitoring results will be submitted to the WDEQ in the annual monitoring report, unless exceedances are detected as outlined in Section 3.1 above. The annual environmental monitoring report will contain the following items:

- A generalized site drawing that illustrates the locations of all methane monitoring probe locations.
- A list of all methane monitoring probes and the probes sampled for each event.
- An explanation of the reason any probes were not sampled during the monitoring event.
- An electronic data set that identifies the methane probe and the corresponding methane results along with any additional parameters monitored (e.g., oxygen).
- Copies of field data sheets.